Component-Driven Regime-Switching Volatility*

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Abstract

We develop a new class of regime-switching volatility models that are characterized by high-dimensional state spaces, parsimonious transition matrices, and ARMA dynamics for the log volatility process. This combination of features is achieved by assuming that we can decompose the Markov chain that describes regime dynamics into a number of two-state component chains that evolve independently through time. Using daily data for S&P 500 index and IBM shares, we show that our component-driven regime-switching (CDRS) models are capable of outperforming GARCH, component GARCH, regime-switching GARCH, and Markov-switching multifractal models in forecasting realized variances out of sample. Interestingly, we find that CDRS models with simple AR(1) dynamics perform well across the board.

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Following the seminal contributions of Hamilton (1989, 1990), regime-switching methods have proven to be highly successful for modeling financial and economic time series. In this paper, we propose a new type of regime-switching volatility specification that we henceforth refer to as a component-driven regime-switching (CDRS) model. The features that distinguish a CDRS model from a conventional regime-switching model include a high-dimensional state space, a parsimonious transition matrix, and a parameterization scheme that implies autoregressive moving average (ARMA) dynamics for the log volatility process. The empirical evidence suggests that CDRS models are capable of outperforming a range of existing models in forecasting the volatility of stock returns out of sample.

Modeling volatility as a regime-switching process has long been a topic of interest in both finance and econometrics. Some recent contributions to the literature include Vo (2009), Wilfling (2009), Bansal et al. (2010), Ichiue and Koyama (2011), Liu et al. (2012), and Maheu et al. (2012). One characteristic shared by most previous studies in this area is that they focus on models with a relatively small number of regimes. Indeed, the simple two-regime model of "bull" and "bear" markets is the most commonly encountered specification in the finance literature. The emphasis on models with a low-dimensional state space appears to stem from both a respect for the value of parsimony, and a desire to attach a specific economic interpretation to each regime.

The Calvet and Fisher (2004) study is a notable exception to this general trend. The authors develop an innovative regime-switching volatility specification — the Markov switching multifractal (MSM) model — that is inspired by research on discrete filters for multifractal diffusions. The MSM model assumes that the volatility of returns can be expressed as a multiplicative function of a number of independent discrete-valued components, each of which follows a very simple stochastic process. In contrast to conventional regime-switching specifications, the MSM model has only four parameters regardless of the dimension of the state space, making it practicable to fit specifications that have hundreds or even thousands of states. Using daily exchange rate data, Calvet and Fisher (2004) demonstrate that an MSM model with ten components, and more than a thousand states, outperforms a set of alternative volatility models along several dimensions.

The CDRS model is similar to the MSM model in that it also assumes volatility can be decomposed into multiple independent components, but the CDRS model employs a different parameterization strategy and therefore has different dynamics implications. The simplest example of the type of model that we consider is one in which the logarithm of volatility is determined by the realization of a two-state Markov chain. Because a two-state Markov chain has a first-order autoregressive (AR(1)) representation, the model implies that the logarithm of volatility follows a discrete-valued AR(1) process. By building on this simple single-component model, we obtain a multi-component model that allows for more complex volatility dynamics while minimizing the potential difficulties that arise from parameter proliferation, overfitting, and other econometric issues.

To develop the multi-component model, we exploit standard results on the aggregation of

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stochastic processes. In particular, we exploit the fact that the sum of independent AR(1) processes is an ARMA process (see, e.g., Engel, 1984). This aggregation property suggests a straightforward way to construct a regime-switching specification in which log volatility displays ARMA dynamics. We simply assume that each component is described by a two-state Markov chain, that each component makes an additive contribution to log volatility, and that each component induces the same change in log volatility when it switches states. The result is a model that is both parsimonious and flexible. For example, an N-component specification with ARMA(2,1) dynamics has only five parameters.

We investigate the performance of the CDRS model using daily returns on the S&P 500 index and IBM shares. As an extension, we briefly examine the performance of a CDRS model for daily realized volatilities of the Deutschemark/U.S. Dollar (DM/$) exchange rate. Our basic strategy is to fit the models via maximum likelihood, and then use Mincer and Zarnowitz (1969) regressions to evaluate the accuracy of the resulting out-of-sample variance forecasts. We follow Andersen and Bollerslev (1998) by using daily realized variances to proxy for the unobserved variances in these regressions.

In addition, we use the Diebold and Mariano (2002) test of equal predictive accuracy to conduct formal pairwise comparisons between the CDRS model and a number of alternative models. We consider the GARCH(1,1) model, which is regarded as a leading benchmark in the volatility modeling literature (see Hansen and Lunde, 2005), and two models whose features overlap to some extent with those of the CDRS model: a two-component GARCH(1,1) (C-GARCH(1,1)) model of the form proposed in Engle and Lee (1999), and a regime-switching GARCH(1,1) (RS-GARCH(1,1)) model of the form proposed in Haas et al. (2004). The tests are based on the mean-squared-error (MSE) loss function, which is robust to noise in the proxy for the unobserved variances (Hansen and Lunde, 2006).

On the whole, the analysis for the S&P 500 index produces the most compelling results. With one component, the CDRS model reduces to a basic two-regime model in which log volatility follows an AR(1) process. The Mincer-Zarnowitz regressions for this specification produce R-squared values of 21.7% with a 1-day forecast horizon and 27.5% with a 22-day horizon (approximately one month of calendar time). Increasing the number of components while maintaining the assumption that the log volatility displays AR(1) dynamics leads to substantial improvements in forecasting performance. For instance, a CDRS model with seven components produces R-squared values of 55.9% and 60.4%. Fitting CDRS models with more complex volatility dynamics leads to further improvements in our measures of in-sample performance. Specifically, models with low-order ARMA dynamics produce smaller values of the Bayesian information criterion (BIC) than models with AR(1) dynamics. In general, however, the reduction in BIC values is not accompanied by increases in the R-squared values for the Mincer-Zarnowitz regressions.

The forecasting performance of the alternative volatility models lends additional perspective to these results. None of the alternative models produces a higher R-squared than a seven-component CDRS model with AR(1) dynamics. Moreover, the Diebold-Mariano tests

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indicate that the performance advantage of the CDRS model is statistically significant in many cases. We generally find that CDRS models with a large number of components perform better than those with a small number of components, and that models with low-order ARMA dynamics perform better than those with high-order ARMA dynamics. The findings for IBM are similar, but the evidence is somewhat weaker with respect to the advantage of the CDRS models over the alternative models in the out-of-sample tests.

Overall our findings suggest that the CDRS model is a promising addition to the regime-switching literature. Like the MSM model of Calvet and Fisher (2004), it is characterized by a high-dimensional state space and a parsimonious transition matrix. However, it has different dynamic implications. Under the CDRS model, the logarithm of volatility has an ARMA representation. We therefore regard the model as a discrete-state-space analog of stochastic volatility models in which volatility, or a simple transformation thereof, follows a standard ARMA process. In view of the good out-of-sample performance of the simple AR(1) specifications in our tests, we believe that the model will prove to be a strong competitor to the MSM model and other regime-switching volatility specifications.

The remainder of the paper is organized as follows. Section 1 discusses component-driven Markov chains, describes our strategy for parameterizing these chains, and provides a detailed description of the proposed class of CDRS models. Section 2 describes the data used for empirical investigation and presents the empirical results. Section 3 provides some brief concluding remarks.

1 CDRS Models

Let the regime for period t be indexed by a discrete random variable $s_t \in \{1, 2, \ldots, K\}$. We consider models in which s_t is generated by an irreducible, ergodic, first-order Markov chain with constant transition probabilities, i.e., the transition from s_t to s_{t+1} is governed by a time-invariant $K \times K$ matrix P with elements $p_{ij} = \Pr(s_{t+1} = j | s_t = i)$. Typically, a first-order Markov chain is specified in terms of a $K \times 1$ vector x_t whose jth element equals one if $s_t = j$ and zero otherwise. The transition equation for x_t is

$$x_{t+1} = P' x_t + e_{t+1}, \qquad t = 1, 2, \ldots, T, \tag{1}$$

where e_{t+1} is a vector martingale difference sequence.

Equation (1) implies that the Markov chain has a first-order vector autoregressive (VAR(1)) representation with $K - 1$ variables. To see this, consider the $K = 2$ case. Using the relation $x_{2t} = 1 - x_{1t}$, we can express the first equation of the bivariate x_t process as

$$x_{1,t+1} = \varpi + \varphi(x_{1,t} - \varpi) + e_{1,t+1}, \qquad t = 1, 2, \ldots, T, \tag{2}$$

where $\varphi = p_{11} - p_{21}$ is the first-order autocorrelation coefficient of $x_{1,t+1}$ and $\varpi = p_{21}/(1 - p_{11} + p_{21})$ is the ergodic probability of the first state, i.e., $\varpi = \mathrm{E}(x_{1,t+1})$. Equation (2) is simply an AR(1) process for $x_{1,t+1}$.

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Now let r_t denote the date t return for some asset, and let $f(r_t|s_t = j, \mathcal{I}_{t-1}; \vartheta)$, where $\mathcal{I}_{t-1} = (r_1, r_2, \ldots, r_{t-1})$ and ϑ is a parameter vector, denote the conditional probability density function (PDF) of r_t given that regime j is operable in period t.[1] In principle, estimation and inference for a given specification of $f(r_t|s_t = j, \mathcal{I}_{t-1}; \vartheta)$ are straightforward. The forecast $x_{t+1|t} = \mathrm{E}(x_{t+1}|\mathcal{I}_t)$ is given in closed form by

$$x_{t+1|t} = \frac{P'(x_{t|t-1} \odot \eta_t)}{\iota'(x_{t|t-1} \odot \eta_t)}, \tag{3}$$

where $\odot$ denotes element-by-element multiplication, $\eta_t = (\eta_{1t}, \ldots, \eta_{kt})'$ is a $K \times 1$ vector with jth element $\eta_{jt} = f(r_t|s_t = j, \mathcal{I}_{t-1}; \vartheta)$, and ι denotes a $K \times 1$ vector of ones. It follows, therefore, that the log likelihood function for the model can be written as

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} \log \iota'(x_{t|t-1} \odot \eta_t), \tag{4}$$

where θ contains the elements of P and ϑ (see, e.g., Hamilton, 1994).

The choice of K is an important aspect of model specification. Occasionally there may be an argument for selecting a particular value of K. For example, in a portfolio optimization setting, a researcher might argue that $K = 2$ is a natural choice for modeling regime-switching volatility because two regimes captures the familiar notion of "bull" and "bear" markets. Although theory provides little guidance on the choice of K, few studies consider models with more than a handful of regimes. The emphasis on models with a small number of regimes appears to be motivated by two considerations. First, models with a small value of K are more parsimonious. Second, specifying a small value of K typically allows the researcher to assign a specific economic interpretation to each regime.

We approach the problem of choosing of K from a different perspective. Parsimony is clearly a concern. However, a regime-switching model for r_t is simply a state-space system in which the states are generated by a discrete-valued VAR(1) process. We see no reason to argue that only small values of K be considered, nor do we see a reason to insist that a specific economic interpretation be attached to each regime. In our view, K should be chosen using a data-based approach that is designed to ensure that the model performs as well as possible in the intended application.

Of course entertaining models with large values of K raises some econometric issues. In general, the dimension of θ grows rapidly with K because P is a $K \times K$ matrix. This leads to concerns about over fitting the data. Moreover, the likelihood function becomes less well behaved as K increases and often displays many local maxima. This makes it more difficult to obtain reliable parameter estimates. We propose a new approach for addressing these issues. The basic idea is to impose a component structure on P to mitigate parameter proliferation,

[1] We assume throughout that the conditional PDF of r_t displays a nontrivial dependence on the value of s_t, i.e., $f(r_t|s_t = j, \mathcal{I}_{t-1}; \vartheta) \neq f(r_t|\mathcal{I}_{t-1}; \vartheta)$. In addition, we model r_t as having a mean of zero in every regime, and estimate the parameters of the models using demeaned returns.

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and parameterize the volatility process such that it displays ARMA dynamics. This yields a parsimonious class of volatility models that includes regime-switching counterparts of many of the stochastic volatility models studied in the literature. We turn now to the details.

1.1 Component-driven Markov chains

Under the proposed class of CDRS models, the Markov chain that governs regime dynamics can be decomposed into a number of simple component chains. Parsimony is achieved by assuming that the component chains evolve independently of one another. To illustrate the general approach, consider a model in which the regime dynamics are driven by $N > 1$ independent two-state Markov chains. It follows from equation (2) that the transition probabilities for a two-state chain satisfy

$$p_{11} = \varphi + \varpi(1 - \varphi), \tag{5}$$
$$p_{22} = 1 - \varpi(1 - \varphi), \tag{6}$$

where $-1 < \varphi < 1$ and $\max\{0, -\varphi/(1-\varphi)\} < \varpi < \min\{1, 1/(1-\varphi)\}$. Hence, we can parameterize the 2^N-state chain that describes the evolution of the regimes by choosing the values of φ and ϖ for each of the N component chains.

Suppose, for example, that $N = 2$. Let $x_t^{(1)}$ and $x_t^{(2)}$ denote the 2×1 vectors that describe the dynamics of the component Markov chains. To construct a four-state Markov chain that describes the joint dynamics of $x_t^{(1)}$ and $x_t^{(2)}$, we simply form s_t via the mapping

$$s_t = \begin{cases} 1 & \text{if } x_{1,t}^{(1)} = 1 \text{ and } x_{1,t}^{(2)} = 1 \\ 2 & \text{if } x_{1,t}^{(1)} = 1 \text{ and } x_{1,t}^{(2)} = 0 \\ 3 & \text{if } x_{1,t}^{(1)} = 0 \text{ and } x_{1,t}^{(2)} = 1 \\ 4 & \text{if } x_{1,t}^{(1)} = 0 \text{ and } x_{1,t}^{(2)} = 0, \end{cases} \tag{7}$$

and specify the transition matrix of the associated x_t process as $P = P^{(1)} \otimes P^{(2)}$, where $P^{(1)}$and $P^{(2)}$ are the transition matrices of $x_t^{(1)}$ and $x_t^{(2)}$.

In general, the transition matrix for a 2^N-state Markov chain is determined by $2^N(2^N - 1)$ transition probabilities. Under a component structure, we can express these probabilities in terms of the $2N$ transition probabilities for the component chains. The transition matrix in our example depends on $p_{11}^{(1)}$, $p_{22}^{(1)}$, $p_{11}^{(2)}$, and $p_{22}^{(2)}$. To specify these probabilities, we set

$$\begin{aligned} p_{11}^{(1)} &= \varphi_1 + \varpi_1(1 - \varphi_1) \\ p_{22}^{(1)} &= 1 - \varpi_1(1 - \varphi_1) \\ p_{11}^{(2)} &= \varphi_2 + \varpi_2(1 - \varphi_2) \\ p_{22}^{(2)} &= 1 - \varpi_2(1 - \varphi_2), \end{aligned} \tag{8}$$

and choose φ_1, ϖ_1, φ_2, and ϖ_2. Parameterizations for $N > 2$ are obtained similarly.

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The two-component parameterization in equation (8) is related to the parameterization of the transition matrix used by Geweke and Amisano (2011) to develop their hierarchical Markov normal mixture (HMNM) model of asset returns. The HMNM model assumes that there are two latent state variables whose dynamics are described by first-order Markov chains with K_1 and K_2 states. The second state variable is assumed to be serially uncorrelated to promote parsimony, but its transition matrix is allowed to depend on the realization of the first state variable. In other words, the HMNM model is based on a K_1K_2-state Markov chain that is driven by two dependent components, one of which is persistent and the other of which is transient. The transition matrix for this chain is determined by $K_1(K_1 + K_2 - 1)$ probabilities. Extending the proposed class of CDRS models to allow for dependent components is straightforward. However, we found that relaxing the independence assumption did little to improve performance in our application.

1.1.1 Implied restrictions on regime dynamics

Specifying a component structure obviously places restrictions on the type of regime dynamics that the model can capture. To see the nature of these restrictions more clearly, assume that the transition matrix in our two-component example can be decomposed as $P = U\Lambda V'$, where $U = (u_1, u_2, u_3, u_4)$ contains the eigenvectors of P in its columns, $\Lambda = \text{diag}\{\lambda_1, \lambda_2, \lambda_3, \lambda_4\}$ contains the eigenvalues of P in decreasing order of magnitude, and $V = (v_1, v_2, v_3, v_4)$ satisfies $U'V = I$.[2] Using standard results for irreducible and ergodic Markov chains, we find that $u_1 = \iota$, $\lambda_1 = 1$ and $v_1 = \pi$, where $\iota = (1, 1, 1, 1)'$ and $\pi = (\pi_1, \pi_2, \pi_3, \pi_4)'$ denotes the vector of ergodic probabilities.

Diagonalizing the transition matrix reveals information about regime dynamics that would not otherwise be apparent. Note that if the value of x_t is known, it follows from the decomposition of P that the conditional k-step-ahead regime probabilities are given by

$$P'^k x_t = \pi + \lambda_2^k v_2 u_2' x_t + \lambda_3^k v_3 u_3' x_t + \lambda_4^k v_4 u_4' x_t, \tag{9}$$

for any integer $k \geq 1$. The second, third and fourth terms on the right side of equation (9) converge to zero as $k \to \infty$, but the rate at which this occurs is different for each term. Specifically, the second term decays more slowly than the third ($|\lambda_2| > |\lambda_3|$), and the third term decays more slowly than the fourth ($|\lambda_3| > |\lambda_4|$).

Because the terms decay at different rates, the eigenvectors of P play a key role in determining the overall dynamics of the chain. If some of these vectors have repeated entries, then we can change the value of x_t without affecting at least one of the terms in equation (9). For example, if the initial three elements of u_2 are identical, then the value of the slowest decaying term is the same regardless of whether the chain is in state 1, 2, or 3 at time t. This corresponds to a scenario in which the low-frequency dynamics of the four-state chain are described by a

[2] If P is not diagonalizable, then similar arguments apply by expressing P in Jordan normal form and analyzing the behavior implied by the Jordan blocks. Models driven by multiple, independent, two-state Markov chains have a diagonalizable transition matrix.

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random variable with only two possible values. In other words, the four-state chain behaves like a two-state chain from this perspective.

It is easy to see that component structures generate this type of behavior. Using the properties of the Kronecker product, we find that $P = P^{(1)} \otimes P^{(2)}$ has the decomposition

$$P = (U^{(1)} \otimes U^{(2)})(\Lambda^{(1)} \otimes \Lambda^{(2)})(V^{(1)} \otimes V^{(2)})', \tag{10}$$

where $P^{(1)} = U^{(1)}\Lambda^{(1)}V^{(1)\prime}$ and $P^{(2)} = U^{(2)}\Lambda^{(2)}V^{(2)\prime}$ are the decompositions for the component chains. Straightforward calculations show that $\lambda_2^{(1)} = \varphi_1$, $\lambda_2^{(2)} = \varphi_2$, $u_2^{(1)} = (-(1-\varpi_1), \varpi_1)'$, and $u_2^{(2)} = (-(1-\varpi_2), \varpi_2)'$. Hence, the eigenvalues of P in decreasing order of magnitude are $\{1, \varphi_1, \varphi_2, \varphi_1\varphi_2\}$, and the eigenvectors are the columns of

$$U = \begin{pmatrix} 1 & -(1-\varpi_1) & -(1-\varpi_2) & (1-\varpi_1)(1-\varpi_2) \\ 1 & -(1-\varpi_1) & \varpi_2 & -(1-\varpi_1)\varpi_2 \\ 1 & \varpi_1 & -(1-\varpi_2) & -\varpi_1(1-\varpi_2) \\ 1 & \varpi_1 & \varpi_2 & \varpi_1\varpi_2 \end{pmatrix}, \tag{11}$$

provided that we label the two chains such that $|\varphi_2| < |\varphi_1|$.

Because the second and third columns of the matrix in equation (11) contain only two distinct entries, there are only two possible realizations of the second and third terms in equation (9). It follows, therefore, that the low-frequency dynamics of the four-regime model are analogous to those of a two-regime model. The behavior that we typically associate with four separate regimes is still present, but it is confined to the high-frequency regime dynamics. In effect, the parsimony of the component structure is achieved by restricting low-frequency dynamics to be characterized by a low number of regimes.

The analysis for models with more than two components is similar. With three components, for example, the low-frequency dynamics are effectively characterized by two regimes, the intermediate-frequency dynamics are effectively characterized by four regimes, and the high-frequency dynamics are characterized by eight regimes. This inverse relation between frequency and number of effective regimes is the fundamental restriction that is imposed by specifying a component structure for the transition matrix.

1.2 Component-driven volatility dynamics

Specifying a component structure for the transition matrix is only the first step in developing a CDRS model. To complete the model, we have to specify how the states of the component chains map to the volatility of returns. Let σ_t denote the volatility of returns in period t, i.e., $\sigma_t^2 = \text{var}(r_t|s_t)$.[3] We assume that each component chain makes an additively separable

[3] Note that $\text{var}(r_t|s_t) = \text{var}(r_t|s_t, \mathcal{I}_{t-1})$ because we assume that the volatility depends solely on the operable regime.

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contribution to $\log \sigma_t$. In particular, we parameterize $\log \sigma_t$ as

$$\log \sigma_t = \psi + \gamma \sum_{i=1}^{N} x_{1,t}^{(i)}, \tag{12}$$

where $\gamma \geq 0$. The value of γ determines the increase in volatility when one of the components switches from the low state ($x_{1,t}^{(i)} = 0$) to the high state ($x_{1,t}^{(i)} = 1$).

The parameterization in equation (12) has a couple of notable features. First, it implies that σ_t has $N + 1$ possible realizations, which is a lot smaller than the number of states of the N-component Markov chain. The reduction in dimensionality is achieved by assuming that the value of γ does not vary across components. Second, it implies that we can view the realization of σ_t as being determined by the outcome of N independent binomial trials in which the probability of a "success" in the ith trial is either $p_{21}^{(i)}$ or $p_{11}^{(i)}$, depending on whether the ith component is in the low or high state in period $t - 1$.

Equation (12) also implies that the dynamics of the volatility process are determined by the dynamics of $\sum_{i=1}^{N} x_{1,t}^{(i)}$. Because each of the component chains follows an AR(1) process, and all of the components are independent, we can invoke standard results to deduce that, in general, the sum of the component chains follows an ARMA(N,$N-1$) process (see, e.g., Engel, 1984). This suggests that the proposed class of CDRS models retains considerable flexibility despite the restrictions imposed to promote parsimony. It also suggests a simple approach for increasing or decreasing the number of components without altering the overall dynamics of volatility.

For example, consider a scenario in which $\varpi_i = \varpi$ and $\varphi_i = \phi$ for $i = 1, 2, \ldots, N$. Because all components are described by the same AR(1) process, it follows that $\sum_{i=1}^{N} x_{1,t}^{(i)}$ is an AR(1) process with autocorrelation parameter ϕ. Accordingly, we obtain a CDRS model with $N+1$ discrete volatility levels in which the log volatility process displays AR(1) dynamics. We will henceforth refer to this specification as a CDRS(N,1,0) model. This designation conveys the two most important features of the model: it has N components, and it implies that the log volatility follows an ARMA(1,0) process. We view the CDRS(N,1,0) model as a natural discrete-state-space analog of the familiar log linear stochastic volatility model of Taylor (1986). It has the same number of parameters as a conventional regime-switching model with two volatility regimes, but it allows for an arbitrary number of volatility levels via the choice of N.

Not surprisingly, the transition matrix for the CDRS(N,1,0) model has a very simple underlying structure. Diagonalizing this $2^N \times 2^N$ matrix reveals that (i) it has $N + 1$ distinct eigenvalues, $\{1, \phi, \phi^2, \ldots, \phi^N\}$, and (ii) the eigenvectors for the ith eigenvalue have i distinct elements. Hence, the autoregressive parameter ϕ determines all of the natural frequencies of the volatility process. The lowest-frequency volatility dynamics, which are analogous to those of a conventional regime-switching volatility model with two regimes, decay according to powers of ϕ. The highest-frequency volatility dynamics, which are analogous to those of a conventional regime-switching volatility model with $N + 1$ regimes, decay according to

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powers of ϕ^N.

Models that display more complex volatility dynamics can be constructed in a straightforward fashion. Consider an alternative parameterization of the component chains in which we set $\varphi_i = \phi_1$ for all $i \leq N-1$ and $\varphi_N = \phi_2$.[4] Under these circumstances, $\sum_{i=1}^{N} x_{1,t}^{(i)}$ can be expressed as the sum of an AR(1) process with autocorrelation parameter ϕ_1 and an AR(1) process with autocorrelation parameter ϕ_2. We therefore obtain a CDRS(N,2,1) model, i.e., a CDRS model in which the log volatility process displays ARMA(2,1) dynamics. More generally, we might consider a range of alternative parameterizations obtained as follows. Let n denote an integer from 1 to N, and let $d = N - n$. Setting $\varphi_i = \phi_1$ for $i \leq d+1$ and $\varphi_i = \phi_{i-d}$ for $i > d+1$ delivers a CDRS(N,n,$n-1$) model.[5]

The choice of n for a given N entails a tradeoff similar to that encountered with standard ARMA models. Using a small n delivers a tightly parameterized specification, which reduces the chances of overfitting the data. But a small n also makes it more likely that the model will fail to accurately describe volatility dynamics. By increasing n, we gain flexibility in modeling the dynamics of volatility across the frequency spectrum. At some point, however, the increased flexibility is likely to become a liability because overfitting the data will impede the out-of-sample forecasting performance of the model.

1.2.1 Extension to realized volatilities

We can use a similar strategy to develop CDRS models for the case in which the data consist of realized volatilities. To illustrate, let rv_t denote the realized variance for period t. If we assume that each component makes an additively separable contribution to $\mu_t = \mathrm{E}(\log rv_t | s_t)$, then the dynamic implications of the resulting CDRS models are similar to those of the CDRS models for returns. In particular, suppose we parameterize μ_t as

$$\mu_t = \psi + \gamma \sum_{i=1}^{N} x_{1,t}^{(i)}, \tag{13}$$

where $\gamma \geq 0$. This produces an N-component model in which μ_t displays ARMA(n, $n-1$) dynamics, where $n \leq N$ is determined by how we parameterize the transition matrix.

Of course we also have to decide how to parameterize the volatility of $\log rv_t$. Let ς_t denote the value of this quantity for period t, i.e., $\varsigma_t^2 = \mathrm{var}(\log rv_t | s_t)$. One possibility is to assume that ς_t is time invariant (independent of s_t), but this is unlikely to be satisfactory because the empirical evidence suggests that ς_t is state dependent (see, e.g., Maheu and McCurdy,

[4] In the interest of parsimony, we assume that ergodic probability ϖ is the same for all components. Relaxing this restriction is straightforward, but has little impact on the empirical results.

[5] Although there are other parameterization strategies for the proposed class of CRDS models that also imply ARMA dynamics for $\log \sigma_t$, this strategy appears to generate the most empirically plausible restrictions for our dataset. For cases in which a preliminary analysis of the data suggests that our strategy is inappropriate, the scope of the model selection problem can be expanded to include the choice of parameterization for each combination of N and n.

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2002). We propose a parameterization for ς_t that mirrors the parameterization for σ_t in the CDRS models for returns, i.e.,

$$\log \varsigma_t = \zeta + \delta \sum_{i=1}^{N} x_{1,t}^{(i)}. \tag{14}$$

Equation (14) implies that any regime shift that affects the value of μ_t also affects the value of ς_t unless $\delta = 0$. We would generally expect to find a positive correlation between the mean and volatility of the log volatility process ($\delta > 0$).

1.3 Comparison set of alternative volatility models

We compare the forecasting performance of the CDRS models to that of a number of alternative volatility models from the literature. The first is a GARCH(1,1) model with t-distributed errors. In particular, we consider a specification of the form

$$r_t = h_t^{1/2} z_t, \tag{15}$$
$$h_t = \omega + \beta h_{t-1} + \alpha r_{t-1}^2, \tag{16}$$

where z_t is an i.i.d. error that has a unit-variance t distribution with ν degrees of freedom. The GARCH(1,1) model is widely used as a performance benchmark in the volatility modeling literature (see Hansen and Lunde, 2005). Specifying a fat-tailed error distribution is intended to allow the GARCH and CDRS models to compete on an equal footing. Although the CDRS model has normally distributed errors conditional on s_t, the distribution of r_t conditional on past returns is a mixture of normals. Hence, the standardized return obtained by dividing r_t by its conditional standard deviation displays excess kurtosis under a CDRS model.

We also consider versions of the C-GARCH model of Engle and Lee (1999) and the RS-GARCH model of Haas et al. (2004). These models are included in the comparison set because they have important elements in common with CDRS models. Under a C-GARCH model, the conditional variance of returns is a mean-reverting process, but the mean is time varying as opposed to constant. We consider a C-GARCH(1,1) specification of the form

$$\begin{aligned} r_t &= h_t^{1/2} z_t, \\ h_t &= m_t + \beta(h_{t-1} - m_{t-1}) + \alpha(r_{t-1}^2 - m_{t-1}), \\ m_t &= \omega + \varphi m_{t-1} + \gamma(r_{t-1}^2 - h_{t-1}), \end{aligned} \tag{17}$$

where z_t is an i.i.d. error that has a unit-variance t distribution with ν degrees of freedom.

Under a RS-GARCH model, on the other hand, the variance in each regime is given by regime-specific GARCH-type recursion. We consider a two-regime RS-GARCH(1,1) specification of the form

$$r_t | s_t = j, \mathcal{I}_{t-1} \sim \mathrm{N}(0, h_{j,t}), \tag{18}$$

where

$$h_{j,t} = \omega_j + \beta_j h_{j,t-1} + \alpha_j r_{t-1}^2 \tag{19}$$

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with $j \in \{1, 2\}$. This is not the only way to formulate a RS-GARCH(1,1) model (see, e.g., Gray, 1996). However, we consider it to be the most natural formulation, and it has the advantage of permitting multi-step ahead variance forecasts in closed form (Haas et al., 2004).

Finally, we consider versions of the MSM model developed by Calvet and Fisher (2004). This is the model from the existing literature that most closely resembles the CDRS model. Under an MSM(N) model, returns are generated by a multiplicative variance-components process of the form

$$r_t = \sigma(c_{1,t}c_{2,t}\cdots c_{N,t})^{1/2} z_t, \tag{20}$$

where the $\{c_{i,t}\}_{i=1}^{N}$ are positive, persistent, mutually-independent random variables with $\mathrm{E}(c_{i,t}) = 1$ for all i and z_t is i.i.d. $N(0,1)$. In particular,

$$c_{i,t} = \begin{cases} c_{i,t-1} \text{ with probability } 1-\nu_i, \\ c_0 \text{ with probability } \nu_i/2, \\ 2-c_0 \text{ with probability } \nu_i/2, \end{cases} \tag{21}$$

where $0 < c_0 < 2$ and the elements of $\nu = (\nu_1, \ldots, \nu_N)'$ satisfy $\log(1-\nu_i) = b^{i-N}\log(1-\nu_N)$ with $0 < \nu_N < 1$ and $b > 1$. Note that ν_N determines the persistence of the least persistent component, and b determines how the persistence varies across the components.

The structural similarities between the MSM and CDRS models are readily apparent. For example, an MSM(2) model implies that the joint dynamics of $c_{1,t}$ and $c_{2,t}$ are described by a four-state Markov chain. To construct this chain, we form s_t via the mapping

$$s_t = \begin{cases} 1 & \text{if } c_{1,t} = c_0,\ c_{2,t} = c_0 \\ 2 & \text{if } c_{1,t} = c_0,\ c_{2,t} = 2-c_0 \\ 3 & \text{if } c_{1,t} = 2-c_0,\ c_{2,t} = c_0 \\ 4 & \text{if } c_{1,t} = 2-c_0,\ c_{2,t} = 2-c_0, \end{cases} \tag{22}$$

and specify the transition probabilities as $p_{ij} = \prod_{i=1}^{2}((1-\nu_i)1_{\{c_{i,t}=c_{i,t-1}\}} + (\nu_i/2))$, where $1_{\{\cdot\}}$ is the indicator function. It is clear from equation (22) that the underlying structure of the MSM(2) model bears a distinct resemblance to that of a CDRS(2,n,$n-1$) model. In each case, the log volatility can be expressed as the sum of two independent components whose dynamics are described by two-state Markov chains.

Despite the structural similarities, it does not necessarily follow that the models have similar dynamic implications. Although the transition matrix for the MSM model is extremely parsimonious, it is difficult to discern the exact nature of the constraints placed on the dynamics of the volatility process. In contrast, the constraints imposed by the CDRS model are very transparent: they imply that the log volatility follows a discrete-valued ARMA process. Given the success of standard ARMA models in a broad range of contexts, we expect the CDRS model to prove quite flexible in capturing volatility dynamics. Whether this flexibil-

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ity translates into better forecasting performance than the more tightly parameterized MSM models is an empirical question.

1.4 Parameter estimation and model selection

The parameter vector for a CDRS(N,n,$n-1$) model is $\theta = (\varpi, \phi_1, \ldots, \phi_n, \psi, \gamma)'$. We estimate θ by maximum likelihood. To construct the log likelihood function, we assume that the demeaned return for period t is drawn from a normal distribution whose variance is determined by the regime. That is,

$$r_t | s_t = j, \mathcal{I}_{t-1} \sim \mathrm{N}(0, \sigma_j^2(\vartheta)), \tag{23}$$

where $\vartheta = (\psi, \gamma)'$ and $\sigma_j(\vartheta)$ denotes the return volatility in regime $j \in (1, 2, \ldots, 2^N)$. The value of this function for a given θ is obtained using equations (3) and (4) with

$$\eta_{jt} = \frac{1}{\sqrt{2\pi}\sigma_j(\vartheta)} \exp\left(-\frac{r_t^2}{2\sigma_j^2(\vartheta)}\right). \tag{24}$$

We maximize the log likelihood using a quasi-Newton method, and compute standard errors using the second-derivative estimate of the information matrix.

To address the model selection issue, we compare the out-of-sample performance of the variance forecasts generated by CRDS models. This requires that we specify a proxy for the unobserved variance of daily returns. We base our analysis on realized variances constructed using high-frequency data. Andersen and Bollerslev (1998) pioneered the use of realized variances for evaluating the forecasting performance of volatility models. Subsequent studies, such as Andersen et al. (2005) and Patton (2011), have refined the procedures and developed methods of inference that are robust to noise in the volatility proxy. We describe our general approach to model selection below and discuss the details in Section 2.

One of the most common strategies used to evaluate the forecasting performance of volatility models is to fit Mincer and Zarnowitz (1969) regressions of the form

$$rv_{t+1} = \kappa_0 + \kappa_1 \hat{\sigma}_{t+1|t}^2 + \epsilon_{t+1}, \tag{25}$$

where rv_{t+1} is the realized variance for period $t+1$ and $\hat{\sigma}_{t+1|t}^2$ is a forecast of the period $t+1$ variance given the period t information set (see, e.g., Fleming and Kirby, 2011). The forecasts are typically constructed using maximum likelihood estimates of the model parameters. Hansen and Lunde (2006) show that using the R-squared from this regression to rank models delivers a ranking that is robust to noise in rv_{t+1}.[6] Unbiased forecasts correspond to $\kappa_0 = 0$ and $\kappa_1 = 1$. We report ordinary least squares (OLS) estimates of these

[6] Some of the studies in the literature specify the regressions in logarithmic terms to reduce the influence of outliers. However, this approach will not, in general, lead to a consistent ranking of the models when the true variances are replaced with realized variances. See Hansen and Lunde (2006) for an in-depth discussion of this issue.

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coefficients, the associated heteroscedasticity and autocorrelation consistent (HAC) standard errors, and the value of the regression R-squared.

To conduct formal pairwise model comparisons, we rely on the Diebold and Mariano (2002) test of equal predictive accuracy. Suppose, for example, that we want to compare the performance of model i to that of model j. Let $\varepsilon_{t+1} = L(\sigma^2_{t+1}, \hat{\sigma}^2_{it+1|t}) - L(\sigma^2_{t+1}, \hat{\sigma}^2_{jt+1|t})$, where $L(\cdot, \cdot)$ is our loss function. The null hypothesis for the Diebold and Mariano (2002) test is $\mathrm{E}(\varepsilon_{t+1}) = 0$. To implement the test, we have to specify both a loss function and a proxy for σ^2_{t+1}. We base our analysis on the MSE loss function, and use rv_{t+1} to proxy for σ^2_{t+1}. The MSE loss function has two features that make it an appealing choice for comparing volatility models. First, it is robust to noise in the proxy for σ^2_{t+1} (Hansen and Lunde, 2006). Second, it is the only robust loss function that treats positive and negative forecast errors symmetrically. We implement the tests for equal predictive accuracy using large-sample t-statistics that are based on HAC standard errors.

2 Data and Empirical Results

The empirical analysis focuses primarily on CDRS models for returns. Although we briefly consider CDRS models for realized volatilities, we leave an in-depth analysis of these models to future research. Our general strategy is to fit the models to the data, and then evaluate the performance of the resulting out-of-sample variance forecasts. We begin with a description of the data, and then follow with a discussion of the empirical results.

2.1 Equity market data

The data used to fit the CDRS models consist of daily returns for the S&P 500 index and IBM shares. Daily returns on IBM shares are drawn from the Center for Research in Security Prices (CRSP) daily stock file. The sample period is January 2, 1926 to January 31, 2001 (20,017 observations). Daily returns for the S&P 500 index are drawn from two sources: Schwert (1990) and the CRSP daily stock file. The Schwert (1990) data cover the interval from January 4, 1928 to July 2, 1962.[7] The CRSP data begin on July 3, 1962. The full sample period is January 4, 1928 to January 31, 2001 (19,433 observations).

The data used to evaluate the forecasting performance of the models are drawn from the on-line data library maintained by the Oxford-Man Institute of Quantitative Finance. They consist of daily realized variances for trading-day returns on the SPDR S&P 500 exchange traded fund and IBM shares.[8] The realized variances are computed using five-minute returns

[7] The returns prior to March 1957 are for the S&P composite index, which contains 90 stocks.

[8] The original source of the realized variances is Noureldin et al. (2011). The high-frequency data used to construct the realized variances are from the Trade and Quote database maintained by the New York Stock Exchange. The procedures used to clean the data and compute the high-frequency returns are described on the Oxford-Man Institute of Quantitative Finance website (http://www.oxford-man.ox.ac.uk).

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with one-minute subsampling to mitigate the impact of microstructure effects. The sample period is February 1, 2001 to December 31, 2009 (2242 observations).

Because the realized variances are based on high-frequency trading-day returns, they are biased estimators of the variances of the daily returns that we use to fit the models. We correct for this bias using a simple rescaling procedure. First we divide the sample variance of the daily returns from the CRSP daily stock file for the period from February 1, 2001 to December 31, 2009 by the mean of the realized variances for the same period. The resulting ratio is 1.743 for the S&P 500 index and 1.536 for IBM shares. Then we rescale the realized variances for each asset by multiplying them by the corresponding ratio. The rescaled data are used to evaluate the forecasting performance of the models.

2.2 Currency market data

For our limited investigation of CDRS models for realized volatility, we use currency market data obtained from Andersen et al. (2003). The data consist of daily realized variances for the DM/$ exchange rate. The realized variances are computed using 30-minute returns. The sample period is December 2, 1986 to June 30, 1999 (3045 observations). We use the data for December 1, 1986 to December 1, 1996 to fit the models, and the data for December 2, 1996 to June 30, 1999 to evaluate the forecasting performance of the models. Because Andersen et al. (2003) use the same estimation and evaluation periods, we can use their results as a baseline for comparison.

2.3 CDRS models for S&P 500 index returns

Table 1 reports the maximum likelihood estimates obtained by fitting selected CDRS models to daily percentage returns on the S&P 500 index. We show results for values of N ranging from one to seven. The parameter estimates in panel A are for CDRS(N,1,0) models. Those in panel B are for CDRS(N,N,$N-1$) models. In each case, setting $N = 1$ delivers the basic two-regime model of "bull" and "bear" markets that has been widely investigated in the finance literature.

The results for the two-regime model are consistent with those reported in prior studies that analyze similar data. If we assume that there are 252 trading days per year, then the estimates of ψ and γ translate into annualized volatilities of 10.7% for the first regime and 33.7% for the second regime. The estimate of ϖ indicates that the second regime is less common than the first regime. It suggests that about one fifth of the days in the sample are characterized by high volatility. The evidence also suggests that the regimes are quite persistent. In particular, the estimate of ϕ_1 is 0.943.

The remainder of the table shows how the results change when we move beyond the simple two-regime model. Panel A shows the effect of increasing N while maintaining the assumption that the log volatility displays AR(1) dynamics. The estimate of ψ, which determines the lowest realization of volatility, initially decreases with N, but it eventually levels off for values

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of $N \geq 5$. The estimate of γ, which determines the increase in log volatility when one of the components switches from the low to high state, decreases monotonically with N. These changes imply a substantial widening of the spread between the lowest and highest volatility levels. With $N = 7$, for example, the lowest realization of volatility is 5.4% per year and the highest is 92.1% per year. The estimate of ϕ_1 also changes with N, rising from 0.969 for $N = 2$ to 0.985 for $N = 7$.

The final row of panel A reports the log likelihood and BIC values for the CDRS$(N, 1, 0)$ models. These statistics point to a substantial improvement in the in-sample fit of the models as we increase the value of N. The two-regime model produces a log likelihood of -25279 and a BIC value of 50,597. With $N = 7$, the log likelihood rises to -24215 and the BIC falls to 48,469. Thus the initial evidence suggests that CDRS models with $N > 1$ provide a significantly better characterization of volatility dynamics than a regime-switching specification with only two volatility regimes.

The results in panel B lend additional perspective to these findings. They illustrate the impact of increasing N while allowing each component to follow a different AR(1) process. In general, the estimates of ψ and γ for a given value of N are similar to the corresponding estimates in panel A. We do observe some divergence in the estimates of γ as N increases, with larger values showing up in panel B. As a consequence, the spread between the lowest and highest levels of volatility is wider than in panel A. With $N = 7$, for example, the lowest and highest realizations of volatility are 4.4% and 133.2% per year.

The estimates of the ϕ_i's point to the presence of highly persistent volatility regimes. Specifically, the estimate of ϕ_1 exceeds 0.99 for $N \geq 2$ and 0.999 for $N \geq 5$. There is also a distinct pattern to the estimates that appears to be consistent with restrictions that imply low-order ARMA dynamics for log volatility. Specifically, for the models with $N \geq 4$, the estimated value of $\phi_i - \phi_{i+1}$ is close to zero for $1 \leq i \leq N - 3$. This suggests that setting $n > 3$ yields little in the way of additional modeling flexibility. On the other hand, the log likelihood and BIC values for the CDRS$(N,N,N-1)$ models indicate that setting $n > 1$ is beneficial. The reduction in the BIC values relative to those reported in panel A ranges from 80 for $N = 2$ to 234 for $N = 6$.

To provide further insights on model selection, we fit CDRS$(N,n,n-1)$ models to S&P 500 returns for values of N ranging from 1 to 7 and all values of $n \leq N$. Table 2 summarizes the model selection results. Panel A reports the log likelihood and BIC values for the estimation period. Panel B reports the R-squared values obtained by regressing the realized variances for the out-of-sample period on the variance forecasts produced by the fitted models. We report results for both 1-day and 22-day forecast horizons. To construct the forecasts for the 22-day horizon, which represents approximately one month of calendar time, we sum the k-step-ahead forecasts of the 1-day return variances for $k = 1$ to $k = 22$. Similarly, we sum the realized variances for the 22 days covered by the forecast horizon to obtain the dependent variable for the regressions. A more detailed description of the regression procedure is provided in Section 2.6.

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First consider the in-sample results in panel A. In general, they are consistent with the pattern noted earlier. The lowest BIC values for the CDRS models with $N \leq 3$ are obtained with $n = N$. Thus the empirical evidence favors setting $n > 1$. However, the lowest BIC values for the models with $N = 4$ and $N = 5$ are obtained with $n = 3$, while those for the models with $N = 6$ and $N = 7$ are obtained with $n = 4$. This suggests that setting n larger than 3 or 4 may be counterproductive. In other words, the evidence from panel A provides little support for CDRS models with high-order ARMA dynamics.

The out-of-sample analysis in panel B tells a similar story. The 1-day and 22-day forecasts from the basic two-regime model produce R-squared values of 21.7% and 27.5%. Increasing N leads to a substantial rise in the R-squared values, underscoring the benefits of allowing for more than two regimes. Moreover, the R-squared values plateau at around 50% to 60% for $N \geq 5$, indicating that the benefits can be achieved with a relatively small number of components. Models with high-order ARMA dynamics generally produce lower R-squared values than models with low-order ARMA dynamics. For instance, the values for the CDRS(7,1,0) model are 55.9% and 60.4%, while those for the CDRS(7,7,6) model are 51.5% and 45.0%. This finding is consistent with the in-sample BIC analysis.

In light of the out-of-sample evidence, the BIC may be a useful criterion for identifying models that can be expected to deliver good forecasts. The R-squared values for the models with $n = 3$ typically exceed those for the other models. Although the models with $n = 4$ fare somewhat worse, the differences are not dramatic. Interestingly, the results indicate that the forecasting performance of models with $n = 1$ and $n = 2$ is comparable to that of models with $n = 3$. Apparently the increase in the R-squared values as we move beyond the basic two-regime model is primarily attributable to increasing N rather than n. Models in which the log volatility displays AR(1) dynamics perform well across the board. Hence, choosing a somewhat more parsimonious model than is indicated by the BIC may be a reasonable approach to model selection.

Table 3 provides a clearer picture of how the properties of models with AR(1) dynamics differ from those of models with low-order ARMA dynamics. Panel A reports estimates of the volatility mass points for the CDRS(7,1,0) and CDRS(7,3,2) models. Panel B reports estimates of the associated switching probabilities. To obtain the estimated switching probability for a given pair of mass points, we form the transition matrix $P = \{p_{ij}\}$ of the composite 128-state Markov chain, and then aggregate across all combinations of i and j that imply transitions between these mass points. [9] For comparison, we also report results for a conventional regime-switching model with eight regimes. Because this model has the

[9] More specifically, we compute the switching probabilities by applying standard aggregation techniques for irreducible Markov chains (see, e.g., Meyer, 1989; Langville and Meyer, 2006). The probability for a given pair of mass points is a weighted average of the elements of P that correspond to a transition between these mass points, where the weights are determined by a subset of the ergodic probabilities of the Markov chain. The ergodic probabilities come into play because we are conditioning only on the initial level of volatility, not on the specific initial regime within the set of regimes that correspond to this level of volatility.

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same number of volatility mass points as the CDRS models, it provides a good basis for assessing the impact of the restrictions imposed by the CDRS models.

Estimation of the conventional model is a challenging task. The likelihood surface appears to have many local optima, making it difficult to ensure that a global maximum is achieved. This is not surprising given that the model has 64 parameters: 56 transition probabilities and 8 volatility mass points. We mitigate this problem by using the estimates from the CDRS(7,1,0) model to compute starting values for the optimizer. Under this approach, the parameter estimates for the conventional model are essentially obtained by fitting the model under the restrictions imposed by the component structure, and then eliminating these restrictions to complete the estimation. Using the estimates from the CDRS(7,3,2) model to compute the starting values yields the same results.

The differences between the volatility mass points for the CDRS(7,1,0) and CDRS(7,3,2) models are relatively small at low to moderate levels of volatility, but the switching probabilities look quite different. The CDRS(7,1,0) models implies a long expected duration at all levels of volatility. Specifically, the estimated probability of remaining at the same volatility level from one day to the next ranges from 93.0% to 96.9%. The CDRS(7,3,2) model, on the other hand, implies much shorter expected durations. The estimated probability of remaining at the same volatility level from one day to the next ranges from 35.9% to 74.7%. The expected durations are shorter because the estimates of ϕ_2 and ϕ_3 are considerably smaller than the estimate of ϕ_1.

A key point on which the two models agree is that many of the switching probabilities are negligible. Of the 64 elements in the matrix of estimated switching probabilities, 31 are zero to four decimal places for the CDRS(7,1,0) model, and 27 are zero to four decimal places for the CDRS(7,3,2) model.[10] The pattern of the estimates for the CDRS(7,1,0) model implies that almost all switches in volatility are between adjacent mass points. No other pattern is possible for this model because the estimate of ϕ_1 is relatively close to one. The pattern of the estimates for the CDRS(7,3,2) model implies that switches between non-adjacent mass points occur slightly more frequently, but such switches are still relatively rare. The most notable difference between the two models is that with ARMA(3,2) dynamics, switches between adjacent mass points are much more frequent.

Although both CDRS models in effect imply that the matrix of switching probabilities is rather sparse, this does not seem to be a consequence of the component structure. The matrix of estimated switching probabilities for the conventional regime-switching model displays a

[10] None of the switching probabilities is exactly zero because the transition matrix has a Kronecker product structure, but the matrix of switching probabilities behaves like a sparse matrix with the same pattern of zeros. There are a number of regime-switching models in the literature that have a sparse transition matrix, such as those based on high-order Markov chains (see, e.g., Raftery, 1985; Berchtold and Raftery, 2002) and those with duration-dependent transition probabilities (see, e.g., Maheu and McCurdy, 2000). However, these models impose sparsity directly, whereas the CDRS models allows for both effectively sparse and dense transition matrices.

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similar sparsity. Specifically, 35 of the estimated switching probabilities in this matrix are zero to four decimal places. The pattern of the estimates for the conventional model is somewhat different from that produced by either CDRS model, but the differences across models are not dramatic. Hence, the CDRS models do not appear to be overly restrictive. This view is bolstered by the results obtained by regressing the realized variances for the out-of-sample period on the variance forecasts produced by the conventional model. The regressions produce an R-squared value of 60% for the 1-day horizon and 55% for the 22-day horizon, which are quite similar to the 56% and 60% values obtained for the CDRS(7,1,0) model.

Overall, the analysis for the S&P 500 index suggests that imposing a component structure on the transition matrix and adopting a parsimonious parameterization of the volatility components is an effective strategy for formulating regime-switching volatility models with a high-dimensional state space. The CDRS models imply a level of sparseness in the matrix of switching probabilities that is broadly consistent with the evidence from a heavily-parameterized, conventional regime-switching specification that is far less econometrically tractable. Moreover, the CDRS models achieve substantial gains in forecasting performance relative to the basic two-regime model that has been widely employed in the finance literature. To assess whether these gains are typical for equity returns, we repeat the model fitting exercise using data for IBM shares.

2.4 CDRS models for IBM returns

Table 4 reports the maximum likelihood parameter estimates obtained by fitting selected CDRS models to daily percentage returns on IBM shares. The estimates of ψ and γ for the basic two-regime model translate into annualized volatilities of 14.5% for the first regime and 38.8% for second regime, and the estimate of ϖ indicates that the second regime is less common than the first regime. In addition, the estimate of ϕ_1, which is 0.900, suggests that the regimes are quite persistent. These results are similar to those in Table 1.

The general pattern of the parameter estimates in Table 4 is also similar to that found in Table 1. First consider the results in panel A for the CDRS(N,1,0) models. As N increases, the estimates of ψ and γ decrease, and the spread between the lowest and highest volatility levels gets wider. With $N = 7$, for example, the lowest realization of volatility is 6.5% per year and the highest is 102.3% per year. We also find that the estimate of ϕ_1 rises from 0.938 for $N = 2$ to 0.957 for $N = 7$. As expected, the log likelihood and BIC values point to a substantial improvement in the in-sample fit of the models as N increases. The two-regime model produces a log likelihood of $-33{,}673$ and a BIC value of 67,386. With $N = 7$, the log likelihood rises to $-32{,}965$ and the BIC falls to 65,970.

Next consider the results in panel B for the CDRS(N,N,$N-1$) models. Once again the estimates of the ϕ_i's point to the presence of highly persistent volatility regimes, and the pattern of the estimates appears to be consistent with restrictions that imply low-order ARMA dynamics for log volatility. In particular, setting $n > 4$ seems to yield little in the

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way of additional modeling flexibility. The reduction in the BIC values relative to those reported in panel A ranges from 228 for $N = 2$ to 375 for $N \geq 6$. All of these findings are consistent with those obtained for the S&P 500 index.

Table 5 presents a more extensive set of model selection results. Panel A shows that the CDRS models with $n = 4$ to $n = 6$ produce the lowest BIC values. Thus the in-sample analysis favors somewhat more complex models for IBM than for the S&P 500 index. Panel B shows that the 1-day and 22-day forecasts from the basic two-regime model produce R-squared values of 23.2% and 31.6%. Increasing N generally leads to a rise in the R-squared values, but not to the same extent as observed in Table 2. There is only one instance in which the R-squared exceeds 40%. Models with high-order ARMA dynamics once again fare worse than models with low-order ARMA dynamics. For instance, the CDRS(7,1,0) models produce R-squared values of 32.7% for the 1-day horizon and 37.4% for the 22-day horizon, while the CDRS(7,7,6) models produce values of 31.1% and 30.0%.

On the whole the results for IBM are qualitatively similar to those for the S&P 500 index. However, they are not as compelling with respect to the advantage of the CDRS models in producing more accurate forecasts than the basic two-regime model. To provide a better frame of reference for evaluating these findings, we turn to the comparison set of alternative volatility models. The performance of the out-of-sample forecasts produced by these models is useful for putting the CDRS evidence in proper perspective. In addition, we can conduct formal tests of equal predictive accuracy to identify statistically significant differences in performance across models.

2.5 Estimation results for alternative volatility models

Table 6 reports maximum likelihood parameter estimates for the set of alternative volatility models. Panel A is for the GARCH(1,1) and C-GARCH(1,1) models, panel B is for the RS-GARCH(1,1) model, and panel C is for the MSM(N) models. We consider two values of N for the MSM models. The MSM(3) model is analyzed because CDRS models with three components appear to perform relatively well. The MSM(7) model is analyzed because it is representative of the specifications found to perform well by Calvet and Fisher (2004). They use values of N up to 10 for daily currency returns, and find that models with larger values of N outperform those with smaller values of N. We use $N = 7$ because this is the maximum number of components considered for the CDRS models, and because the performance differences documented by Calvet and Fisher (2004) are not statistically significant for $N = 7$ versus $N = 10$ (see their Table 4).

The GARCH(1,1) results in panel A are typical of the literature. The estimates of α and β, which are 0.072 and 0.923 for the S&P 500 index, and 0.087 and 0.908 for IBM, are indicative of strong volatility persistence. The estimates of ν point to conditionally fat-tailed errors. Similarly, the C-GARCH(1,1) results are qualitatively consistent with those of prior studies that fit this model to daily equity returns. The estimate of φ is 0.969 for the S&P 500 index and 1 for IBM (rounded to three decimal places). These values are suggestive of a

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slowly mean-reverting, long-run component in volatility. The estimates of α and β are 0.019 and 0.977 for the S&P 500 index, and 0.103 and 0.764 for IBM, indicating that the second component of volatility is also highly persistent. In each case, the C-GARCH model produces a lower BIC value than the GARCH model. The difference in the BIC values is 10 for the S&P 500 index and 125 for IBM.

The RS-GARCH(1,1) results in panel B are similar to those reported by Haas et al. (2004) for daily currency returns. The estimates of α_1, β_1, and p_{11} are 0.346, 0.882, and 0.234 for the S&P 500 index, and 0.156, 0.943, and 0.492 for IBM. Note that $\alpha_1 + \beta_1 > 1$ for each asset, so the results imply that the first regime is characterized by nonstationary behavior. The estimates of α_2, β_2, and p_{22} are 0.043, 0.934, and 0.888 for the S&P 500 index, and 0.028, 0.950, and 0.915 for IBM. It is easy to verify that the full set of parameter estimates satisfy the eigenvalue restriction necessary for the process to be covariance stationary (see Haas et al., 2004, for details).

The MSM(3) and MSM(7) results in panel C are in line with those reported by Calvet and Fisher (2004). The estimates of σ and c_0 for MSM(3) model imply that the lowest and highest realizations of volatility are 6.4% per year and 47.1% per year for the S&P 500 index, and 8.1% per year and 57.6% per year for IBM. The estimates of ν_3 and b imply that the volatility components are highly persistent. In the case of the third component, which displays the least persistence, the estimated probability that $c_{3,t} = c_{3,t-1}$ is 0.948 for the S&P 500 index and 0.842 for IBM.

As with the CDRS models, increasing the number of components leads to a widening of the spread between the minimum and maximum volatility levels. The estimates of σ and c_0 for the MSM(7) model imply that the lowest and highest realizations of volatility are 3.1% and 65.6% per year for the S&P 500 index, and 2.9% and 80.4% per year for IBM. In addition, increasing N causes the estimated persistence of the least persistence component to decline substantially. The estimated probability that $c_{7,t} = c_{7,t-1}$ is 0.611 for the S&P 500 index and 0.618 for IBM.

2.6 Performance comparisons using alternative volatility models

Table 7 reports the results of Mincer-Zarnowitz regressions for the alternative models and a selected set of CDRS models. We include the CDRS(3,1,0) model in the table because our analysis suggests that at least three components are needed to generate accurate variance forecasts, and specifications with AR(1) dynamics appear to perform well as a general proposition. Similarly, we include the CDRS(7,1,0) and CDRS(7,3,2) models because of their generally good performance, and to clearly illustrate the effect of specification changes along two dimensions: increasing N without altering overall volatility dynamics, and introducing more complex volatility dynamics while holding N constant.

To construct Table 7, we use OLS to fit simple linear regression models of the form

$$rv_{t+1,H} = \kappa_0 + \kappa_1 \hat{\sigma}^2_{t+1,H} + \epsilon_{t+1,H}, \tag{26}$$

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where $H \in \{1, 5, 22\}$ is the forecast horizon in days, $rv_{t+1,H} = \sum_{k=1}^{H} rv_{t+k}$ is the H-day realized variance obtained by summing the daily realized variances for days $t+1$ to $t+H$, and $\hat{\sigma}^2_{t+1,H} = \sum_{k=1}^{H} \hat{\sigma}^2_{t+k|t}$ is the forecast of the H-day return variance obtained by summing the k-step-ahead forecasts of the 1-day return variances for $k = 1$ to $k = H$. The value of t starts at T, the time index of the last day of the sample used to construct the parameter estimates for the volatility models, and ends at $T + M$, where $M = 2242 - H$ is determined by the number of available H-day realized variance observations. To make the results easier to interpret, we divide $rv_{t+1,H}$ and $\hat{\sigma}^2_{t+1,H}$ by H before estimating the regression coefficients. This has no effect on the estimates of κ_1 or the R-squared values, but it rescales the estimates of κ_0 such that they can be directly compared for the three different forecast horizons.

The results for the S&P 500 index are shown in panel A. For the GARCH(1,1) model, the estimates of κ_0 and κ_1 are 0.26 and 0.89 for the 1-day forecasts, but we cannot reject $\kappa_0 = 0$ or $\kappa_1 = 1$ at the 5% level. Hence, the observed bias is not statistically significant. As the horizon increases, the estimate of κ_0 rises, the estimate of κ_1 falls, and the bias becomes statistically significant. We reject both $\kappa_0 = 0$ and $\kappa_1 = 1$ at 5% for the 5-day and 22-day forecasts. The performance of the C-GARCH(1,1) model is better. The estimates of κ_0 and κ_1 are 0.13 and 1.02 for the 1-day horizon. There are small changes in the estimates as the horizon increases, but there is no statistically significant bias at any horizon. The RS-GARCH(1,1) model performs about the same as the GARCH(1,1) model, although the estimates of κ_1 for the 5- and 22-day horizons are closer to one.

The behavior of the coefficient estimates for the MSM(3) and MSM(7) models is similar to that for the GARCH(1,1) and C-GARCH(1,1) models, but the biases are in the opposite direction. For the 1-day forecasts, the estimates of κ_0 are -0.49 and -0.59, the estimates of κ_1 are 1.32 and 1.40, and we can reject $\kappa_0 = 0$ at 5% in both cases. As the horizon increases, the estimates of κ_0 rise, the estimates of κ_1 fall, and the biases become statistically insignificant. In general, the evidence suggests that the MSM models do not perform as well as the GARCH(1,1) and C-GARCH(1,1) models, especially for the 22-day horizon. The R-squared values for this horizon are 42% for the MSM(3) model, 43% for the MSM(7) model, and 55% for both the GARCH(1,1) and C-GARCH(1,1) models.

The CDRS models prove to be stronger competitors in this regard. The CDRS(3,1,0) model displays some evidence of statistically significant bias at the 1-day horizon. However, the CDRS(7,1,0) and CDRS(7,3,2) models display none. For these two models, the estimates of κ_0 are -0.03 and 0.06 for the 1-day forecasts, and the estimates of κ_1 are 1.10 and 0.97. The changes in the estimates as the horizon increases are relatively small, and the R-squared values indicate that these models perform well relative to the GARCH(1,1), C-GARCH(1,1), and RS-GARCH(1,1) models. Indeed, the CDRS(7,1,0) model delivers the highest R-squared value of any model for each horizon: 56% for the 1-day forecasts, 67% for the 5-day forecasts, and 60% for the 22-day forecasts.

Turning to the results for IBM in panel B, we find that the estimates of κ_0 are greater than 0 and the estimates of κ_1 are less than 1 for both the GARCH(1,1) and C-GARCH(1,1)

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models. However, the biases are generally not statistically significant. The RS-GARCH(1,1) model again performs about the same as the GARCH(1,1) model, but produces estimates of κ_1 that are closer to one. Although the MSM(3) model displays statistically significant bias at the 1-day horizon, the MSM(7) model performs well, with no statistically significant bias at any horizon. In addition, the R-squared values for the MSM models are higher than those for the GARCH(1,1) and C-GARCH(1,1) models at every horizon. Thus the GARCH(1,1) and C-GARCH(1,1) models do not enjoy a clear advantage over the MSM models for IBM. The R-squared values for the CDRS(7,1,0) and CDRS(7,3,2) models are comparable to those for the MSM models, and neither displays any statistically significant bias. Moreover, the CDRS(3,1,0) model delivers the highest R-squared value for each horizon.

If we consider the results in Table 7 as a whole, it is apparent that the forecasting performance of CDRS models compares quite favorably with that of the alternative volatility models. In addition, the evidence with respect to the relative performance of the various CDRS models parallels that from the GARCH literature. A number of studies show the simple GARCH(1,1) model, which implies an ARMA(1,1) representation for the squared demeaned returns, is tough to beat in terms of forecasting performance. Similarly, we find that the simple CDRS(7,1,0) model, which implies AR(1) dynamics for the log volatility, performs well relative to more complex CDRS models, and also relative to the GARCH, C-GARCH, RS-GARCH, and MSM models.

One possible caveat regarding the evidence in Table 7 is that the parameter estimates used to construct the forecasts are based on data that span more than 70 years. If the data generating process displays any form of instability (e.g., occasional structural breaks), then using such a long sample of returns for parameter estimation might adversely affect the forecasting performance of the models. To shed light on whether this is a potential concern, Table 8 replicates the analysis of Table 7 using parameter estimates obtained by fitting the models to data for July 3, 1962 to January 31, 2001. By using only post-war data to fit the models, we ensure that the variance forecasts are unaffected by the atypical conditions that prevailed in U.S. equity markets during both the Great Depression and World War II. We specify July 3, 1962 as the starting date for the reduced sample because this is the first date that the S&P 500 index return is available from the CRSP daily stock file.

In general, Table 8 looks very much like Table 7. Consider the results for the S&P 500 index. It is still the case that the CDRS(7,1,0) model delivers the highest R-squared value of any model for each horizon. The only difference is that these values are slightly lower than those in Table 7: 52% for the 1-day forecasts, 64% for the 5-day forecasts, and 56% for the 22-day forecasts. Similarly, the CDRS(3,1,0) model delivers the highest R-squared value at each horizon for IBM. There are some minor changes in the findings for various models, e.g., the performance of the RS-GARCH(1,1) model for the S&P 500 index deteriorates a bit relative to that of the GARCH(1,1) and C-GARCH(1,1) models. However, using a reduced sample for parameter estimation does not have a meaningful impact on our overall assessment of the forecasting performance of the different models.

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Table 9 addresses the statistical significance of the performance differences across models. Specifically, it reports the results of pairwise Diebold-Mariano tests of equal predictive accuracy using the MSE loss function. For the pair of models in row i and column j of the table, we compute the time series of daily loss differentials $\{\varepsilon_{t+1,H}\}_{t=T}^{T+M}$ as

$$\varepsilon_{t+1,H} = (rv_{t+1,H} - \hat{\sigma}^2_{it+1,H})^2 - (rv_{t+1,H} - \hat{\sigma}^2_{jt+1,H})^2,$$

where $H \in \{1, 22\}$ is the forecast horizon in days and $M = 2242 - H$. The test is based on the t-statistic for the hypothesis $\mathrm{E}(\varepsilon_{t+1,H}) = 0$. A negative (positive) value of the t-statistic indicates that the model in the row i generates a smaller (larger) average loss than the model in column j. Panel A is for the S&P 500 index, and panel B is for IBM.

The t-statistics in panel A convey much the same message as the Mincer-Zarnowitz regressions. The GARCH(1,1), C-GARCH(1,1), and RS-GARCH(1,1) models have smaller average losses than the MSM models. However, the differences are not statistically significant. The CDRS models generally produce smaller average losses than the alternative models, and in many cases these differences are statistically significant. The performance of the CDRS(7,1,0) model again stands out in this regard. For the 1-day horizon, it generates t-statistics of 1.64, 2.59, and 2.27 with respect to the GARCH(1,1), C-GARCH(1,1) ,and RS-GARCH(1,1) models, and 1.63 and 1.67 with respect to the MSM(3) and MSM(7) models. Although the t-statistics for the 22-day horizon are somewhat smaller, the general pattern is the same.

The results in panel B for IBM are also consistent the Mincer-Zarnowitz regressions. Once again we find that the CDRS models generally produce smaller average losses than the other models, and the t-statistics indicate that a number of the differences are statistically significant. However, the performance advantage of the CDRS(7,1,0) model is not as clear cut as in panel A, largely because of the improved relative performance of the MSM models. The evidence also suggests that the CDRS(3,1,0) model performs better than in panel A, especially relative to the C-GARCH(1,1) model.

These findings, and the results of Mincer-Zarnowitz regressions, suggest that CDRS models offer a promising new approach to modeling stochastic volatility. It is especially interesting that the CDRS models fare well in comparison to MSM models. As noted earlier, CDRS and MSM models share a number of structural similarities, but differ with respect to the approach used to parameterize the transition matrix. The approached used in MSM models is motivated by research on discrete filters for multifractal diffusions. In contrast, CDRS models build on standard results regarding sums of independent linear processes to obtain parameterizations that imply ARMA dynamics for the log volatility. Although more research is needed to draw firm conclusions, the empirical analysis suggests that the latter approach may ultimately prove to be more successful.

2.7 CDRS models for DM/$ realized volatilities

To develop preliminary evidence on the potential merits of CDRS models for realized volatilities, we fit the CDRS(3,1,0), CDRS(7,1,0), and CDRS(7,3,2) models to daily DM/$ exchange

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rate data. Table 10 presents the results. Panel A reports the maximum likelihood parameter estimates for the December 1, 1986 to December 1, 1996 sample period. Panel B reports the results of Mincer-Zarnowitz regressions using data for December 2, 1996 to June 30, 1999.

Most of the properties of the parameter estimates in panel A are familiar from the analysis of CDRS models for returns. The estimates of the ϕ_i's point to strong volatility persistence, and the estimates of ψ and γ imply that the spread between the lowest and highest volatility levels is wider with $N = 7$ than with $N = 3$. In addition, the estimates of δ are positive and statistically significant for all three models. This indicates that the conditional volatility of rv_t is time varying, and that changes in the conditional mean and volatility of rv_t are positively correlated. Based on the BIC values, the CDRS(7,3,2) delivers the best in-sample performance among the three specifications.

The results of the Mincer-Zarnowitz regressions in panel B reveal little difference in performance between the CDRS(3,1,0) and CDRS(7,1,0) models. The models have similar estimates of κ_0 and κ_1 at each horizon, and both display evidence of statistically significant bias at the 5-day horizon. They also produce similar R-squared values at each horizon: 13% and 14% for the 1-day forecasts, 15% and 17% for the 5-day forecasts, and 15% and 16% for the 22-day forecasts. The CDRS(7,3,2) model performs slightly worse, showing statistically significant bias at both the 5-day and 22-day horizons, but it manages to produce the same R-squared value as the CDRS(7,1,0) model in each case.

To lend some perspective to these findings, Andersen et al. (2003) report that fitting a GARCH(1,1) model to daily DM/$ returns for the same sample period produces 1-day forecasts with an R-squared value of 9.6%, and that fitting a long-memory VAR model to daily realized volatilities for the DM/$, Yen/$ and Yen/DM exchange rates for this period produces 1-day forecasts with an R-squared value of 24.9%. Thus the CDRS models perform better than the GARCH(1,1) model, but fall substantially short of the performance of the long-memory VAR model. In light of these results, it might be interesting to investigate whether multivariate CDRS model can improve on the performance of the univariate models examined in Table 10.

3 Conclusion

Modeling volatility as a regime-switching process has been the subject of considerable research in finance and econometrics. Traditionally, however, the focus has been on models with a low-dimensional state space. This recently changed with introduction of the MSM model by Calvet and Fisher (2004), which allows the Markov chain that describes regime dynamics to have hundreds or even thousands of states. We develop a new type of regime-switching volatility specification — the CDRS model — that provides a similar level of flexibility with respect to specifying the evolution of the regimes. Under the proposed model, the logarithm of volatility follows a discrete-valued ARMA process. Hence, we regard the model as a regime-switching analog of conventional stochastic volatility models in which volatility, or

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a simple transformation thereof, has an ARMA representation.

The empirical investigation suggests that the CDRS model is a promising addition to the regime-switching literature. We find that the model is capable of outperforming GARCH, component GARCH, regime-switching GARCH, and MSM models in forecasting realized variances out of sample. In addition, the analysis reveals that CDRS specifications with AR(1) dynamics generally deliver the best overall performance. Our findings in this regard cast the results of Calvet and Fisher (2004) in an interesting light. Although their analysis indicates that the MSM model performs well in forecasting exchange rate volatility, this may not lend as much support to the multifractal framework as it initially seems. Perhaps the volatility process actually displays simple dynamics, and the success of the MSM model is attributable to its approximating characteristics. In either case, our results vis-à-vis the MSM model show that the CDRS model is worthy of further investigation.

We should also point out that the framework developed in the paper can be generalized along several dimensions. One possibility is to consider multivariate CDRS models. In a multivariate setting, a model with multiple independent components is suggestive of a latent factor structure. It might therefore be useful to formulate a multivariate CDRS model that has both common and idiosyncratic volatility components. This could be accomplished in a straightforward fashion. Another possibility is to develop a class of Markov switching regression models in which the regression coefficients display ARMA dynamics. We leave these and other generalizations to future research.

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Table 1
Maximum likelihood estimates of CDRS parameters for the S&P 500 index

Panel A: CDRS(N,1,0) models

	$N=1$		$N=2$		$N=3$		$N=4$		$N=5$		$N=6$		$N=7$	
Parameter	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE
ψ	-0.3936	0.0078	-0.5928	0.0137	-0.7709	0.0207	-0.8149	0.0204	-1.0898	0.0322	-1.0858	0.0267	-1.0772	0.0274
γ	1.1468	0.0141	0.7612	0.0094	0.6165	0.0092	0.5424	0.0103	0.4841	0.0095	0.4256	0.0106	0.4051	0.0091
ϖ	0.2002	0.0174	0.2582	0.0194	0.2998	0.0198	0.2733	0.0175	0.3586	0.0208	0.3380	0.0188	0.3010	0.0165
ϕ_1	0.9426	0.0053	0.9690	0.0028	0.9771	0.0019	0.9791	0.0017	0.9849	0.0014	0.9851	0.0013	0.9852	0.0013
LL & BIC	-25279	50597	-24641	49322	-24378	48795	-24287	48614	-24244	48527	-24224	48487	-24215	48469

Panel B: CDRS(N,N,$N-1$) models

	$N=1$		$N=2$		$N=3$		$N=4$		$N=5$		$N=6$		$N=7$	
Parameter	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE
ψ	-0.3936	0.0078	-0.6230	0.0168	-0.8227	0.0191	-0.8740	0.0223	-0.9811	0.0342	-1.3156	0.0511	-1.2719	0.0324
γ	1.1468	0.0141	0.7817	0.0101	0.6628	0.0102	0.5822	0.0096	0.5169	0.0104	0.5007	0.0135	0.4856	0.0132
ϖ	0.2002	0.0174	0.2477	0.0201	0.2572	0.0217	0.2224	0.0174	0.2631	0.0264	0.2257	0.0245	0.2010	0.0222
ϕ_1	0.9426	0.0053	0.9933	0.0014	0.9978	0.0007	0.9989	0.0006	0.9994	0.0004	0.9998	0.0001	0.9997	0.0004
ϕ_2			0.9095	0.0103	0.9865	0.0029	0.9949	0.0015	0.9942	0.0041	0.9988	0.0008	0.9996	0.0004
ϕ_3					0.8334	0.0218	0.9674	0.0054	0.9935	0.0047	0.9958	0.0016	0.9959	0.0028
ϕ_4							0.6561	0.0677	0.9365	0.0101	0.9851	0.0047	0.9959	0.0028
ϕ_5									0.2237	0.0940	0.9200	0.0151	0.9726	0.0099
ϕ_6											0.1814	0.0881	0.9089	0.0224
ϕ_7													0.1414	0.0939
LL & BIC	-25279	50597	-24596	49242	-24277	48613	-24169	48408	-24127	48333	-24082	48253	-24076	48250

The table summarizes the results of fitting selected CDRS models to daily percentage returns on the S&P 500 index. Under a CDRS(N,n,$n-1$) model, volatility dynamics are governed by N independent, 2-state Markov chains, $\{x_t^{(i)}\}_{i=1}^N$. We parameterize the transition probabilities for $x_t^{(i)}$ as $p_{11}^{(i)} = \varphi_i + \varpi(1-\varphi_i)$ and $p_{22}^{(i)} = 1 - \varpi(1-\varphi_i)$, where $\varphi_i = \phi_1$ for $i \leq d+1$ and $\varphi_i = \phi_{i-d}$ for $i > d+1$ with $d = N - n$, and consider models of the form

$$r_t | s_t = j, \mathcal{I}_{t-1} \sim N(0, \sigma_t^2), \quad t = 1, 2, \ldots, T,$$
$$\log \sigma_t = \psi + \gamma \textstyle\sum_{i=1}^N x_{1,t}^{(i)},$$

where $j \in (1, 2, \ldots, 2^N)$ and $\mathcal{I}_{t-1} = (r_1, r_2, \ldots, r_{t-1})$. We report the maximum likelihood parameter estimates (Est), the associated standard errors (SE), the value of the log likelihood (LL), and the Bayesian information criterion, $BIC = -2LL + m \log T$ where m is the number of parameters. The sample period is January 4, 1928 to January 31, 2001 (T = 19,433 observations).

Table 2
Model selection criteria for the S&P 500 index

Panel A: Measures of in-sample fit

	$N=1$		$N=2$		$N=3$		$N=4$		$N=5$		$N=6$		$N=7$	
Model	LL	BIC	LL	BIC	LL	BIC	LL	BIC	LL	BIC	LL	BIC	LL	BIC
CDRS(N,1,0)	-25279	50597	-24641	49322	-24378	48795	-24287	48614	-24244	48527	-24224	48487	-24215	48469
CDRS(N,2,1)			-24596	49242	-24282	48614	-24189	48426	-24150	48349	-24141	48331	-24122	48292
CDRS(N,3,2)					-24277	48613	-24171	48400	-24130	48320	-24093	48246	-24088	48236
CDRS(N,4,3)							-24169	48408	-24129	48327	-24086	48241	-24080	48230
CDRS(N,5,4)									-24127	48333	-24084	48246	-24079	48237
CDRS(N,6,5)											-24082	48253	-24076	48241
CDRS(N,7,6)													-24076	48250

Panel B: R-squared of out-of-sample forecasts

	$N=1$		$N=2$		$N=3$		$N=4$		$N=5$		$N=6$		$N=7$	
Model	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day
CDRS(N,1,0)	0.217	0.275	0.371	0.440	0.432	0.513	0.561	0.618	0.550	0.612	0.550	0.585	0.559	0.604
CDRS(N,2,1)			0.378	0.400	0.457	0.421	0.540	0.495	0.525	0.484	0.523	0.486	0.517	0.527
CDRS(N,3,2)					0.468	0.430	0.516	0.460	0.529	0.516	0.522	0.499	0.504	0.474
CDRS(N,4,3)							0.516	0.458	0.533	0.516	0.540	0.502	0.510	0.452
CDRS(N,5,4)									0.529	0.505	0.537	0.489	0.526	0.470
CDRS(N,6,5)											0.546	0.508	0.515	0.450
CDRS(N,7,6)													0.515	0.450

The table summarizes the in-sample fit and out-of-sample forecasting performance of CDRS models for the S&P 500 index. Under a CDRS(N,n,$n-1$) model, volatility dynamics are governed by N independent, 2-state Markov chains, $\{x_t^{(i)}\}_{i=1}^N$. We parameterize the transition probabilities for $x_t^{(i)}$ as $p_{11}^{(i)} = \varphi_i + \varpi(1-\varphi_i)$ and $p_{22}^{(i)} = 1 - \varpi(1-\varphi_i)$, where $\varphi_i = \phi_1$ for $i \leq d+1$ and $\varphi_i = \phi_{i-d}$ for $i > d+1$ with $d = N-n$, and consider models of the form

$$r_t | s_t = j, \mathcal{I}_{t-1} \sim N(0, \sigma_t^2), \quad t = 1, 2, \ldots, T,$$
$$\log \sigma_t = \psi + \gamma \textstyle\sum_{i=1}^N x_{1,t}^{(i)},$$

where $j \in (1, 2, \ldots, 2^N)$ and $\mathcal{I}_{t-1} = (r_1, r_2, \ldots, r_{t-1})$. Panel A reports the value of the log likelihood function (LL) and the Bayesian information criterion ($BIC = -2LL + m \log T$ where m is the number of parameters) using data for January 4, 1928 to January 31, 2001 (T =19,433 daily percentage returns). Panel B reports the R-squared for an OLS regression of the realized variances for February 1, 2001 to December 31, 2009 (2242 observations) on the variance forecasts produced by fitting the model using the data for January 4, 1928 to January 31, 2001.

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Table 3
Volatility mass points and associated switching probabilities for the S&P 500 index

Panel A: Volatility mass points

CDRS(7,1,0) model	92.2	61.5	41.0	27.3	18.2	12.2	8.1	5.4
CDRS(7,3,2) model	122.2	76.0	47.2	29.4	18.3	11.3	7.1	4.4
Conventional model	81.5	35.9	30.4	20.4	13.0	11.5	6.0	5.7

Panel B: Switching probabilities

CDRS(7,1,0) model

	92.2	61.5	41.0	27.3	18.2	12.2	8.1	5.4
92.2	0.9300	0.0678	0.0021	0.0000	0.0000	0.0000	0.0000	0.0000
61.5	0.0042	0.9358	0.0585	0.0015	0.0000	0.0000	0.0000	0.0000
41.0	0.0000	0.0084	0.9415	0.0490	0.0010	0.0000	0.0000	0.0000
27.3	0.0000	0.0001	0.0127	0.9472	0.0395	0.0006	0.0000	0.0000
18.2	0.0000	0.0000	0.0001	0.0170	0.9528	0.0298	0.0003	0.0000
12.2	0.0000	0.0000	0.0000	0.0002	0.0214	0.9584	0.0200	0.0001
8.1	0.0000	0.0000	0.0000	0.0000	0.0003	0.0258	0.9639	0.0100
5.4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0004	0.0303	0.9693

CDRS(7,3,2) model

	122.3	76.0	47.3	29.4	18.3	11.4	7.1	4.4
122.3	0.3587	0.6011	0.0398	0.0004	0.0000	0.0000	0.0000	0.0000
76.0	0.0325	0.4163	0.5224	0.0285	0.0002	0.0000	0.0000	0.0000
47.3	0.0003	0.0660	0.4733	0.4412	0.0191	0.0001	0.0000	0.0000
29.4	0.0000	0.0008	0.1003	0.5295	0.3577	0.0115	0.0000	0.0000
18.3	0.0000	0.0000	0.0016	0.1356	0.5851	0.2719	0.0058	0.0000
11.4	0.0000	0.0000	0.0000	0.0028	0.1717	0.6399	0.1836	0.0019
7.1	0.0000	0.0000	0.0000	0.0000	0.0041	0.2088	0.6940	0.0930
4.4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0058	0.2468	0.7474

Conventional model

	81.5	35.9	30.4	20.4	13.0	11.5	6.0	5.7
81.5	0.8443	0.1556	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
35.9	0.0252	0.9466	0.0000	0.0282	0.0000	0.0000	0.0000	0.0000
30.4	0.0000	0.0080	0.5777	0.0000	0.2855	0.1287	0.0000	0.0000
20.4	0.0000	0.0122	0.0000	0.9701	0.0177	0.0000	0.0000	0.0000
13.0	0.0000	0.0000	0.0186	0.0085	0.9691	0.0035	0.0003	0.0000
11.5	0.0000	0.0000	0.0119	0.0000	0.0025	0.6181	0.3674	0.0001
6.0	0.0000	0.0000	0.0000	0.0000	0.0006	0.4843	0.5078	0.0072
5.7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0186	0.9812

The table reports maximum likelihood estimates obtained by fitting a CDRS(7,1,0) model, a CDRS(7,3,2) model, and a conventional regime-switching model with eight regimes to daily percentage returns on the S&P 500 index. Panel A reports the estimated value of each volatility mass point in annualized percentage terms. Panel B reports the estimated probability of switching from the row mass point to the column mass point in adjacent time periods. The CDRS models assume that the dynamics of volatility are governed by 7 independent, 2-state Markov chains, $\{x_t^{(i)}\}_{i=1}^{7}$. We parameterize the transition probabilities for $x_t^{(i)}$ as $p_{11}^{(i)} = \varphi_i + \varpi(1-\varphi_i)$ and $p_{22}^{(i)} = 1 - \varpi(1-\varphi_i)$, where $\varphi_i = \phi_1$ for $i \leq d+1$ and $\varphi_i = \phi_{i-d}$ for $i > d+1$ with $d = 6$ and $d = 4$, and specify the models as

$$r_t | s_t = j, \mathcal{I}_{t-1} \sim N(0, \sigma_t^2), \quad t = 1, 2, \ldots, T,$$
$$\log \sigma_t = \psi + \gamma \textstyle\sum_{i=1}^{N} x_{1,t}^{(i)},$$

where $j \in (1, 2, \ldots, 128)$ and $\mathcal{I}_{t-1} = (r_1, r_2, \ldots, r_{t-1})$. To obtain the switching probability for a given pair of volatility mass points, we form the transition matrix $P = \{p_{ij}\}$ of the composite 128-state Markov chain, and then aggregate across all combinations of i and j that imply transitions between these mass points.

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Table 4
Maximum likelihood estimates of CDRS parameters for IBM shares

Panel A: CDRS(N,1,0) models

	$N=1$		$N=2$		$N=3$		$N=4$		$N=5$		$N=6$		$N=7$	
Parameter	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE
ψ	-0.0866	0.0106	-0.3057	0.0182	-0.5059	0.0295	-0.5897	0.0405	-0.6921	0.0383	-0.7090	0.0372	-0.8888	0.0732
γ	0.9812	0.0133	0.7254	0.0110	0.6038	0.0109	0.5351	0.0115	0.4745	0.0112	0.4376	0.0107	0.3932	0.0096
ϖ	0.2857	0.0162	0.3228	0.0171	0.3630	0.0182	0.3448	0.0216	0.3541	0.0197	0.3267	0.0174	0.3764	0.0258
ϕ_1	0.8999	0.0081	0.9375	0.0047	0.9502	0.0038	0.9538	0.0036	0.9565	0.0037	0.9577	0.0038	0.9565	0.0040
LL & BIC	-33673	67386	-33180	66399	-33035	66110	-32992	66023	-32973	65986	-32966	65972	-32965	65970

Panel B: CDRS(N,N,$N-1$) models

	$N=1$		$N=2$		$N=3$		$N=4$		$N=5$		$N=6$		$N=7$	
Parameter	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE	Est	SE
ψ	-0.0866	0.0106	-0.3058	0.0176	-0.5352	0.0231	-0.6551	0.0311	-0.6956	0.0303	-0.7065	0.0382	-0.7319	0.0438
γ	0.9812	0.0133	0.7752	0.0115	0.6463	0.0103	0.5617	0.0104	0.5056	0.0101	0.4696	0.0127	0.4419	0.0134
ϖ	0.2857	0.0162	0.2716	0.0182	0.2930	0.0184	0.2905	0.0216	0.2879	0.0208	0.2570	0.0235	0.2358	0.0251
ϕ_1	0.8999	0.0081	0.9902	0.0018	0.9976	0.0007	0.9992	0.0004	0.9997	0.0003	0.9997	0.0002	0.9996	0.0004
ϕ_2			0.7712	0.0200	0.9628	0.0052	0.9853	0.0027	0.9945	0.0020	0.9975	0.0013	0.9988	0.0009
ϕ_3					0.6243	0.0366	0.8959	0.0173	0.9749	0.0052	0.9841	0.0041	0.9882	0.0040
ϕ_4							0.4217	0.0628	0.7869	0.0406	0.9170	0.0242	0.9621	0.0125
ϕ_5									0.4011	0.0909	0.6522	0.0803	0.7747	0.1423
ϕ_6											0.3246	0.1330	0.6606	0.2254
ϕ_7													0.2350	0.1506
LL & BIC	-33673	67386	-33061	66171	-32864	65787	-32799	65668	-32767	65614	-32754	65597	-32748	65595

The table summarizes the results of fitting selected CDRS models to daily percentage returns on IBM shares. Under a CDRS(N,n,$n-1$) model, volatility dynamics are governed by N independent, 2-state Markov chains, $\{x_t^{(i)}\}_{i=1}^N$. We parameterize the transition probabilities for $x_t^{(i)}$ as $p_{11}^{(i)} = \varphi_i + \varpi(1-\varphi_i)$ and $p_{22}^{(i)} = 1 - \varpi(1-\varphi_i)$, where $\varphi_i = \phi_1$ for $i \leq d+1$ and $\varphi_i = \phi_{i-d}$ for $i > d+1$ with $d = N-n$, and consider models of the form

$$r_t | s_t = j, \mathcal{I}_{t-1} \sim N(0, \sigma_t^2), \quad t = 1, 2, \ldots, T,$$
$$\log \sigma_t = \psi + \gamma \textstyle\sum_{i=1}^N x_{1,t}^{(i)},$$

where $j \in (1, 2, \ldots, 2^N)$ and $\mathcal{I}_{t-1} = (r_1, r_2, \ldots, r_{t-1})$. We report the maximum likelihood parameter estimates (Est), the associated standard errors (SE), the value of the log likelihood (LL), and the Bayesian information criterion, $BIC = -2LL + m \log T$ where m is the number of parameters. The sample period is January 2, 1926 to January 31, 2001 (T = 20,017 observations).

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Table 5
Model selection criteria for IBM shares

Panel A: Measures of in-sample fit

	$N=1$		$N=2$		$N=3$		$N=4$		$N=5$		$N=6$		$N=7$	
Model	LL	BIC	LL	BIC	LL	BIC	LL	BIC	LL	BIC	LL	BIC	LL	BIC
CDRS(N,1,0)	-33673	67386	-33180	66399	-33035	66110	-32992	66023	-32973	65986	-32966	65972	-32965	65970
CDRS(N,2,1)			-33061	66171	-32891	65831	-32854	65758	-32847	65743	-32852	65753	-32859	65767
CDRS(N,3,2)					-32864	65787	-32813	65685	-32787	65634	-32786	65631	-32789	65637
CDRS(N,4,3)							-32799	65668	-32771	65612	-32766	65601	-32765	65600
CDRS(N,5,4)									-32767	65614	-32755	65589	-32755	65588
CDRS(N,6,5)											-32754	65597	-32748	65585
CDRS(N,7,6)													-32748	65595

Panel B: R-squared of out-of-sample forecasts

	$N=1$		$N=2$		$N=3$		$N=4$		$N=5$		$N=6$		$N=7$	
Model	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day
CDRS(N,1,0)	0.232	0.316	0.337	0.387	0.368	0.398	0.330	0.372	0.325	0.373	0.324	0.371	0.327	0.374
CDRS(N,2,1)			0.332	0.304	0.355	0.388	0.358	0.392	0.338	0.381	0.332	0.373	0.329	0.369
CDRS(N,3,2)					0.360	0.406	0.357	0.366	0.337	0.347	0.339	0.357	0.336	0.353
CDRS(N,4,3)							0.359	0.365	0.329	0.331	0.323	0.319	0.328	0.333
CDRS(N,5,4)									0.318	0.313	0.306	0.291	0.313	0.301
CDRS(N,6,5)											0.306	0.289	0.313	0.302
CDRS(N,7,6)													0.311	0.300

The table presents measures of the in-sample fit and out-of-sample forecasting performance of CDRS models for IBM. Under a CDRS(N,n,$n-1$) model, volatility dynamics are governed by N independent, 2-state Markov chains, $\{x_t^{(i)}\}_{i=1}^N$. We parameterize the transition probabilities for $x_t^{(i)}$ as $p_{11}^{(i)} = \varphi_i + \varpi(1-\varphi_i)$ and $p_{22}^{(i)} = 1 - \varpi(1-\varphi_i)$, where $\varphi_i = \phi_1$ for $i \leq d+1$ and $\varphi_i = \phi_{i-d}$ for $i > d+1$ with $d = N-n$, and consider models of the form

$$r_t | s_t = j, \mathcal{I}_{t-1} \sim N(0, \sigma_t^2), \quad t = 1, 2, \ldots, T,$$
$$\log \sigma_t = \psi + \gamma \textstyle\sum_{i=1}^N x_{1,t}^{(i)},$$

where $j \in (1, 2, \ldots, 2^N)$ and $\mathcal{I}_{t-1} = (r_1, r_2, \ldots, r_{t-1})$. Panel A reports the value of the log likelihood function (LL) and the Bayesian information criterion ($BIC = -2LL + m \log T$ where m is the number of parameters) using data for January 2, 1926 to January 31, 2001 (T = 20,017 daily percentage returns). Panel B reports the R-squared for an OLS regression of the realized variances for February 1, 2001 to December 31, 2009 (2242 observations) on the variance forecasts produced by fitting the model using the data for January 2, 1926 to January 31, 2001.

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Table 6
Maximum likelihood estimates of GARCH and MSM parameters

Panel A: GARCH(1,1) and C-GARCH(1,1) models

	BIC	ω	SE	β	SE	α	SE	ν	SE	φ	SE	γ	SE
S&P 500 index													
GARCH(1,1)	48279	0.007	0.001	0.923	0.004	0.072	0.004	5.935	0.237				
C-GARCH(1,1)	48269	0.021	0.003	0.977	0.004	0.019	0.004	6.305	0.250	0.969	0.005	0.056	0.005
IBM shares													
GARCH(1,1)	65760	0.025	0.003	0.908	0.007	0.087	0.007	4.841	0.162				
C-GARCH(1,1)	65635	0.005	0.001	0.764	0.033	0.103	0.010	4.939	0.171	1.000	0.001	0.031	0.004

Panel B: RS-GARCH(1,1) model

		Regime 1								Regime 2							
	BIC	ω_1	SE	β_1	SE	α_1	SE	p_{11}	SE	ω_2	SE	β_2	SE	α_2	SE	p_{22}	SE
S&P 500 index	48445	0.039	0.017	0.882	0.022	0.346	0.066	0.234	0.045	0.004	0.001	0.934	0.005	0.043	0.004	0.888	0.017
IBM shares	65876	0.074	0.021	0.943	0.010	0.156	0.027	0.492	0.042	0.006	0.001	0.950	0.006	0.028	0.004	0.915	0.009

Panel C: MSM(N) models

	BIC	σ	SE	b	SE	ν_N	SE	c_0	SE
S&P 500 index									
MSM(3)	48683	1.491	0.026	11.863	1.813	0.105	0.012	0.419	0.007
MSM(7)	48308	1.244	0.028	6.284	0.609	0.779	0.074	0.591	0.007
IBM shares									
MSM(3)	65854	1.837	0.041	12.340	1.450	0.317	0.035	0.425	0.007
MSM(7)	65620	1.403	0.047	5.922	0.533	0.764	0.064	0.557	0.009

The table summarizes the results of fitting the GARCH and MSM models to daily percentage returns on the S&P 500 index and IBM shares. It reports the maximum likelihood parameter estimates, the associated standard errors (SE), and the Bayesian information criterion, $BIC = -2LL + m \log T$ where LL is the value of the log likelihood and m is the number of parameters. The sample period is January 4, 1928 to January 31, 2001 (19,433 observations) for the S&P 500 index, and January 2, 1926 to January 31, 2001 (20,017 observations) for IBM. See the text for details on modeling assumptions and model specification.

Table 7
Mincer-Zarnowitz regressions

Panel A: S&P 500 index

	1-Day Horizon					5-Day Horizon					22-Day Horizon				
Model	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2
GARCH(1,1)	0.26	0.15	0.89	0.11	0.51	0.35	0.15	0.84	0.12	0.62	0.56	0.19	0.74	0.15	0.55
C-GARCH(1,1)	0.13	0.15	1.02	0.12	0.53	0.20	0.16	0.99	0.13	0.63	0.36	0.20	0.95	0.18	0.55
RS-GARCH(1,1)	0.30	0.14	0.85	0.11	0.49	0.36	0.16	0.91	0.13	0.61	0.56	0.20	0.97	0.20	0.53
MSM(3)	-0.49	0.22	1.32	0.17	0.41	-0.48	0.28	1.31	0.22	0.50	-0.35	0.40	1.21	0.31	0.42
MSM(7)	-0.59	0.25	1.40	0.18	0.49	-0.51	0.29	1.36	0.22	0.55	-0.23	0.38	1.22	0.31	0.43
CDRS(3,1,0)	-0.41	0.22	1.42	0.19	0.43	-0.41	0.27	1.44	0.24	0.54	-0.57	0.46	1.64	0.41	0.51
CDRS(7,1,0)	-0.03	0.16	1.10	0.12	0.56	0.01	0.17	1.10	0.14	0.67	0.01	0.23	1.17	0.21	0.60
CDRS(7,3,2)	0.06	0.19	0.97	0.13	0.50	0.15	0.21	0.93	0.16	0.58	0.35	0.28	0.85	0.22	0.47

Panel B: IBM shares

	1-Day Horizon					5-Day Horizon					22-Day Horizon				
Model	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2
GARCH(1,1)	0.42	0.32	0.82	0.14	0.29	0.52	0.37	0.79	0.16	0.37	0.77	0.45	0.69	0.19	0.32
C-GARCH(1,1)	0.36	0.32	0.81	0.13	0.27	0.37	0.31	0.79	0.14	0.34	0.60	0.34	0.69	0.13	0.27
RS-GARCH(1,1)	0.29	0.30	0.88	0.14	0.28	0.37	0.31	0.93	0.16	0.35	0.66	0.41	0.97	0.23	0.30
MSM(3)	-0.52	0.25	1.13	0.12	0.34	-0.59	0.31	1.14	0.15	0.43	-0.50	0.40	1.09	0.20	0.36
MSM(7)	-0.23	0.22	0.98	0.10	0.33	-0.20	0.24	0.96	0.12	0.40	0.02	0.33	0.87	0.15	0.33
CDRS(3,1,0)	-0.37	0.28	1.20	0.14	0.37	-0.56	0.35	1.29	0.18	0.48	-1.41	0.63	1.68	0.33	0.40
CDRS(7,1,0)	0.02	0.35	1.05	0.16	0.33	-0.11	0.40	1.12	0.20	0.42	-0.86	0.61	1.46	0.32	0.37
CDRS(7,3,2)	0.14	0.29	0.86	0.12	0.34	0.17	0.28	0.84	0.13	0.42	0.45	0.34	0.74	0.15	0.35

The table summarizes the results of regressing the realized variances for February 1, 2001 to December 31, 2009 (2242 observations) on the variance forecasts produced by fitting the models to the daily percentage returns for January 4, 1928 to January 31, 2001 for the S&P 500 index, and January 2, 1926 to January 31, 2001 for IBM shares. The regressions are of the form

$$rv_{t+1,H} = \kappa_0 + \kappa_1 \hat{\sigma}^2_{t+1,H} + \epsilon_{t+1,H},$$

where $H \in \{1, 5, 22\}$ is the forecast horizon, $rv_{t+1,H} = \sum_{k=1}^{H} rv_{t+k}$ is the H-day realized variance obtained by summing the daily realized variances for days $t+1$ to $t+H$, and $\hat{\sigma}^2_{t+1,H} = \sum_{k=1}^{H} \hat{\sigma}^2_{t+k|t}$ is the forecast of the H-day return variance obtained by summing the k-step-ahead forecasts of the 1-day return variances for $k = 1$ to $k = H$. We report OLS estimates of κ_0 and κ_1, robust standard errors based on Newey-West weights (SE), and the regression R-squared. The Newey-West weights are constructed using a lag length of 5 for the 1-day horizon, 10 for the 5-day horizon, and 30 for the 22-day horizon.

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Table 8
Mincer-Zarnowitz regressions using alternative parameter estimates

Panel A: S&P 500 index

	1-Day Horizon					5-Day Horizon					22-Day Horizon				
Model	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2
GARCH(1,1)	0.28	0.15	0.88	0.11	0.50	0.36	0.16	0.84	0.12	0.61	0.59	0.20	0.73	0.15	0.54
C-GARCH(1,1)	0.17	0.16	0.98	0.12	0.51	0.25	0.17	0.95	0.14	0.62	0.44	0.21	0.88	0.18	0.54
RS-GARCH(1,1)	0.28	0.15	0.92	0.12	0.44	1.11	0.08	0.89	0.14	0.56	1.30	0.14	0.80	0.18	0.50
MSM(3)	-0.71	0.23	1.66	0.21	0.34	-0.71	0.29	1.65	0.27	0.43	-0.68	0.46	1.61	0.41	0.39
MSM(7)	-0.74	0.27	1.61	0.21	0.46	-0.69	0.32	1.60	0.26	0.54	-0.45	0.42	1.47	0.37	0.44
CDRS(3,1,0)	-0.67	0.22	1.78	0.23	0.36	-0.65	0.28	1.80	0.29	0.45	-0.68	0.46	1.93	0.49	0.43
CDRS(7,1,0)	0.14	0.16	1.02	0.13	0.52	0.19	0.18	1.01	0.14	0.64	0.27	0.24	1.04	0.22	0.56
CDRS(7,3,2)	-0.12	0.18	1.14	0.14	0.51	-0.03	0.19	1.10	0.15	0.60	0.16	0.24	1.04	0.22	0.51

Panel B: IBM shares

	1-Day Horizon					5-Day Horizon					22-Day Horizon				
Model	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2
GARCH(1,1)	0.18	0.25	0.92	0.12	0.26	0.27	0.30	0.89	0.15	0.34	0.55	0.41	0.80	0.18	0.30
C-GARCH(1,1)	0.06	0.30	0.95	0.14	0.27	0.13	0.33	0.93	0.15	0.35	0.38	0.40	0.84	0.18	0.29
RS-GARCH(1,1)	-0.05	0.35	0.99	0.15	0.24	-0.25	0.37	1.06	0.17	0.33	-0.11	0.52	1.01	0.22	0.29
MSM(3)	-0.95	0.29	1.34	0.14	0.32	-0.97	0.35	1.34	0.18	0.41	-0.79	0.45	1.25	0.23	0.34
MSM(7)	-0.62	0.24	1.22	0.12	0.31	-0.54	0.26	1.19	0.14	0.39	-0.22	0.35	1.07	0.18	0.32
CDRS(3,1,0)	-0.42	0.28	1.21	0.14	0.36	-0.46	0.34	1.24	0.18	0.47	-0.79	0.52	1.40	0.28	0.40
CDRS(7,1,0)	0.16	0.36	0.97	0.16	0.32	0.16	0.40	0.97	0.19	0.41	-0.02	0.50	1.06	0.25	0.36
CDRS(7,3,2)	0.04	0.26	0.98	0.13	0.33	0.13	0.28	0.95	0.14	0.41	0.45	0.34	0.84	0.17	0.35

The table summarizes the results of regressing the realized variances for February 1, 2001 to December 31, 2009 (2242 observations) on the variance forecasts produced by fitting the models to the daily percentage returns for July 3, 1962 to January 31, 2001. The regressions are of the form

$$rv_{t+1,H} = \kappa_0 + \kappa_1 \hat{\sigma}^2_{t+1,H} + \epsilon_{t+1,H},$$

where $H \in \{1, 5, 22\}$ is the forecast horizon, $rv_{t+1,H} = \sum_{k=1}^{H} rv_{t+k}$ is the H-day realized variance obtained by summing the daily realized variances for days $t+1$ to $t+H$, and $\hat{\sigma}^2_{t+1,H} = \sum_{k=1}^{H} \hat{\sigma}^2_{t+k|t}$ is the forecast of the H-day return variance obtained by summing the k-step-ahead forecasts of the 1-day return variances for $k = 1$ to $k = H$. We report OLS estimates of κ_0 and κ_1, robust standard errors based on Newey-West weights (SE), and the regression R-squared. The Newey-West weights are constructed using a lag length of 5 for the 1-day horizon, 10 for the 5-day horizon, and 30 for the 22-day horizon.

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Table 9
Pairwise tests of equal predictive accuracy

Panel A: S&P 500 index

	C-GARCH(1,1)		RS-GARCH(1,1)		MSM(3)		MSM(7)		CDRS(3,1,0)		CDRS(7,1,0)		CDRS(7,3,2)	
Model	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day
GARCH(1,1)	0.96	0.62	-2.12	0.16	-1.03	-0.42	-0.76	-0.35	-1.02	-0.35	1.64	0.89	0.01	-0.34
C-GARCH(1,1)			-1.94	-1.29	-1.36	-0.95	-1.25	-0.91	-1.34	-0.79	2.59	1.53	-1.04	-1.50
RS-GARCH(1,1)					-0.87	-0.75	-0.47	-0.72	-0.86	-0.61	2.27	1.89	1.13	-0.73
MSM(3)							1.45	0.33	0.08	0.42	1.63	1.29	1.22	0.43
MSM(7)									-1.41	-0.04	1.67	1.27	0.98	0.34
CDRS(3,1,0)											1.61	1.10	1.20	0.32
CDRS(7,1,0)													-1.92	-1.64

Panel B: IBM shares

	C-GARCH(1,1)		RS-GARCH(1,1)		MSM(3)		MSM(7)		CDRS(3,1,0)		CDRS(7,1,0)		CDRS(7,3,2)	
Model	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day	1 Day	22 Day
GARCH(1,1)	-0.84	-0.35	0.15	0.22	1.07	0.71	0.97	0.44	1.45	0.45	1.47	0.52	1.87	0.61
C-GARCH(1,1)			1.13	1.14	1.58	1.90	1.60	1.97	1.96	1.22	2.62	1.40	2.50	1.19
RS-GARCH(1,1)					1.45	0.97	1.53	0.43	1.87	0.67	2.06	0.79	1.64	0.17
MSM(3)							-0.61	-1.28	1.30	-0.64	-0.39	-0.61	-0.28	-0.63
MSM(7)									1.18	0.25	0.10	0.37	0.02	-0.12
CDRS(3,1,0)											-1.16	0.35	-0.73	-0.22
CDRS(7,1,0)													-0.08	-0.29

The table reports large-sample t-statistics for pairwise Diebold-Mariano tests of equal predictive accuracy under MSE loss for 1-day and 22-day forecast horizons. For the pair of models in row i and column j, the loss differential for period $t+1$ is computed as

$$\varepsilon_{t+1,H} = (rv_{t+1,H} - \hat{\sigma}^2_{it+1,H})^2 - (rv_{t+1,H} - \hat{\sigma}^2_{jt+1,H})^2,$$

where $H \in \{1, 22\}$ is the forecast horizon, $rv_{t+1,H} = \sum_{k=1}^{H} rv_{t+k}$ is the H-day realized variance obtained by summing the daily realized variances for days $t+1$ to $t+H$, $\hat{\sigma}^2_{it+1,H} = \sum_{k=1}^{H} \hat{\sigma}^2_{it+k|t}$ is the forecast of the H-day return variance produced by the model in row i (obtained by summing the k-step-ahead forecasts of the 1-day return variances for $k = 1$ to $k = H$), and $\hat{\sigma}^2_{jt+1,H} = \sum_{k=1}^{H} \hat{\sigma}^2_{jt+k|t}$ is the forecast of the H-day return variance produced by the model in column j. The null hypothesis for the test is $\mathrm{E}(\varepsilon_{t+1,H}) = 0$. The t-statistics are based on robust standard errors that are constructed using Newey-West weights. The lag length for the Newey-West weights is 10 for the 1-day horizon and 40 for the 22-day horizon. A positive (negative) t-statistic indicates that the row i model performs worse (better) than the column j model. The parameters used to construct the forecasts are estimated using daily percentage returns for January 4, 1928 to January 31, 2001 for the S&P 500 index and January 2, 1926 to January 31, 2001 for IBM shares. The sample period for the tests is February 1, 2001 to December 31, 2009 (2242 observations).

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Table 10
Results of fitting CDRS models to DM/$ realized variances

Panel A: Parameter estimates

	CDRS(3,1,0)		CDRS(7,1,0)		CDRS(7,3,2)	
Parameter	Est	SE	Est	SE	Est	SE
ψ	-1.0366	0.0230	-1.2421	0.0342	-1.4044	0.0378
γ	0.3642	0.0099	0.2483	0.0098	0.3006	0.0122
ζ	-1.4102	0.0420	-1.5404	0.0558	-2.0232	0.1132
δ	0.0473	0.0222	0.0574	0.0155	0.1275	0.0245
ϖ	0.4397	0.0357	0.3944	0.0284	0.3495	0.0366
ϕ_1	0.9332	0.0098	0.9414	0.0095	0.9691	0.0056
ϕ_2					0.3624	0.1246
ϕ_3					0.3624	0.1246
LL & BIC	-593	1233	-569	1186	-535	1132

Panel B: Mincer-Zarnowitz regressions

	1-Day Horizon					5-Day Horizon					22-Day Horizon				
Model	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2	κ_0	SE	κ_1	SE	R^2
CDRS(3,1,0)	0.05	0.05	0.90	0.15	0.13	0.12	0.06	0.70	0.16	0.15	0.09	0.09	0.73	0.25	0.15
CDRS(7,1,0)	0.05	0.05	0.92	0.15	0.14	0.12	0.05	0.71	0.15	0.17	0.11	0.08	0.69	0.22	0.16
CDRS(7,3,2)	0.04	0.05	0.89	0.15	0.14	0.11	0.05	0.69	0.15	0.17	0.16	0.06	0.57	0.18	0.16

The table summarizes the results of fitting selected CDRS models to daily realized variances for the DM/$ exchange rate. Under a CDRS(N,n,$n-1$) model, the dynamics of realized volatility are governed by N independent, 2-state Markov chains, $\{x_t^{(i)}\}_{i=1}^N$. We parameterize the transition probabilities for $x_t^{(i)}$ as $p_{11}^{(i)} = \varphi_i + \varpi(1-\varphi_i)$ and $p_{22}^{(i)} = 1 - \varpi(1-\varphi_i)$, where $\varphi_i = \phi_1$ for $i \leq d+1$ and $\varphi_i = \phi_{i-d}$ for $i > d+1$ with $d = N-n$, and consider models of the form

$$\begin{gathered}\log rv_t | s_t = j, \mathcal{I}_{t-1} \sim N(\mu_t, \varsigma_t^2), \quad t = 1, 2, \ldots, T, \\ \mu_t = \psi + \gamma \textstyle\sum_{i=1}^N x_{1,t}^{(i)}, \\ \log \varsigma_t = \zeta + \delta \textstyle\sum_{i=1}^N x_{1,t}^{(i)},\end{gathered}$$

where $j \in (1, 2, \ldots, 2^N)$ and $\mathcal{I}_{t-1} = (rv_1, rv_2, \ldots, rv_{t-1})$. Panel A reports the maximum likelihood parameter estimates (Est), the associated standard errors (SE), the value of the log likelihood (LL), and the Bayesian information criterion, $BIC = -2LL + m \log T$ where m is the number of parameters. The sample period is December 1, 1986 to December 1, 1996 ($T = 2449$ observations). Panel B presents the results of regressing the realized variances for December 2, 1996 to June 30, 1999 (596 observations) on the forecasts obtained using the parameter estimates in panel A. The regressions are of the form

$$rv_{t+1,H} = \kappa_0 + \kappa_1 \hat{\sigma}_{t+1,H}^2 + \epsilon_{t+1,H},$$

where $H \in \{1, 5, 22\}$ is the forecast horizon, $rv_{t+1,H} = \sum_{k=1}^H rv_{t+k}$ is the H-day realized variance obtained by summing the daily realized variances for days $t+1$ to $t+H$, and $\hat{\sigma}_{t+1,H}^2 = \sum_{k=1}^H \hat{\sigma}_{t+k|t}^2$ is the forecast of the H-day return variance obtained by summing the k-step-ahead forecasts of the 1-day return variances for $k = 1$ to $k = H$. We report OLS estimates of the intercept and slope coefficients, robust standard errors based on Newey-West weights (SE), and the R-squared for the regression (R^2). The Newey-West weights are constructed using a lag length of 5 for the 1-day horizon, 10 for the 5-day horizon, and 30 for the 22-day horizon.

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